ARITHMETIC ENVELOPES ON THE CIRCLE: AN INTRODUCTION

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Abstract. We study envelopes of families of chords generated by arithmetic maps on the unit circle, which we call arithmetic envelopes. For the doubling and tripling maps these are the classical cardioid and nephroid, which are also caustics of a circular mirror; more generally, the map $x \mapsto kx \bmod 1$ produces an epicycloid with $k-1$ cusps. We then change the coordinate system on the circle by labelling the odd integers with dyadic rationals, which identifies them with the Prüfer 2-group, and we recall its group law from earlier work and describe it by a simple rule on binary digits. In these coordinates multiplication by 3 becomes a piecewise linear circle homeomorphism, conjugate to the rotation by $\log_2 3$, and its chords leave a region in the centre of the circle free. We prove that this region, observed by Henkel and called the convexoid, is convex and bounded by a segment of a diameter and by arcs of two epicycloids with cusps at $1/5$ and $-1/7$. The pedal curve of the same family is heart-shaped and lies on the algebraic curves $x = \rho\, T_5(\rho)$ and $x = \rho\, T_7(\rho)$ with $\rho^2 = x^2 + y^2$, where T_n denotes the n-th Chebyshev polynomial. We compute these and further envelopes explicitly with Gröbner bases and resultants, and we include code for all computations; an appendix discusses the tropicalization of the resulting curves. We close with questions on how such pictures might help to visualize problems such as Furstenberg's $\times 2$ $\times 3$ conjecture.

1. Introduction

Place points on a circle, label them by numbers, and join every label x to the label of $g(x)$ for some arithmetic map g. For the doubling map this "times-table" picture shows a cardioid, for the tripling map a nephroid. Neither curve is drawn by any single line; each is the *envelope* of the whole family of lines. The purpose of this paper is to use such envelopes, which we call *arithmetic envelopes*, as a small geometric toolbox for arithmetic dynamical systems on the circle group and on some of its discrete subgroups, such as the Prüfer 2-group.

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The paper is mainly expository: most individual curves are classical, and our aim is to present them in a coherent, computational and visual way. The dyadic coordinate system on the circle and the name "dreamcatcher" for the resulting families of lines were introduced by I. J. Aberkane [1, 2]. In [2] the same coordinate system, together with its Prüfer 2-group structure, was used to visualize known solutions of sums of three cubes and of cubic quadruples ("diophantine imaging"). A three-dimensional rendering of the underlying binary and ternary trees on a cone was developed in Henkel's master's thesis [3], which also observed the empty, caustic-like region left by the chords of multiplication by 3. Here we focus on the geometry of the envelopes themselves. Our main new result, Theorem 1, determines this region exactly; most other results are classical, and we include them with short proofs to keep the paper self-contained.

The thread of the paper is as follows. Section 2 sets up envelopes of chord families and gives a formula for the point at which each line touches its envelope. Section 3 applies it to the maps $x \mapsto kx \bmod 1$, whose envelopes are epicycloids and, for $k = 2, 3$, classical caustics; this is where Furstenberg's $\times 2$ $\times 3$ problem provides the motivation. Section 4 changes the labelling of the circle: odd integers are placed according to their binary digits, which identifies them with the Prüfer 2-group. In these coordinates, multiplication by 3 is conjugate to the irrational rotation by $\log_2 3$, so that Furstenberg's principle reappears, and Section 5 proves that its chords leave free a convex region bounded by a segment and two epicycloid arcs (Theorem 1). Section 6 derives the polynomial equations of the heart-shaped pedal curve of the same family. Section 7 treats implicit equations of epicycloids and a non-linear map. We conclude with an outlook in Section 8. Appendix A discusses the tropicalization of the heart-shaped curve, Appendix B a piecewise linear labelling that reproduces the cardioid, Appendix C collects code, explicit formulas and the computer-assisted step of the proof of Theorem 1, and Appendix D recalls algebraic background.

2. Envelopes of chord families

Throughout, we identify $\mathbb{R}^2$ with $\mathbb{C}$ and write $\mathbb{T}$ for the unit circle.

Definition 1 (Arithmetic envelope, dreamcatcher)**.** *Let $g\colon X \to X$ be an arithmetic map and let $f\colon X \to \mathbb{T}$ embed the set X into the unit circle.*

- *The family of lines connecting $f(x)$ and $f(g(x))$ for all $x \in X$ is called the* ***dreamcatcher of*** *g* ***in base*** *f.*
- *If X is an interval and the lines depend smoothly on x, a curve that is tangent to every line of the dreamcatcher (see equation 1) is called the* ***arithmetic envelope*** *of g in base f.*

The term dreamcatcher follows [1, 2].

Consider a family of curves $\{C_t\}$ given implicitly by $F(x, y, t) = 0$. A curve that is tangent to C_t for every t is called an *envelope* of the family; it is given by the system

$$F(x, y, t) = 0, \qquad \frac{\partial F}{\partial t}(x, y, t) = 0, \tag{1}$$

from which its equation is obtained by eliminating the parameter t [4, p. 213]. For a dreamcatcher the curves C_t are the lines through two points $P_1(t)$ and $P_2(t)$ of the unit circle, given by the two-point form [5, p. 22]

$$(y - y_{P_1})(x_{P_2} - x_{P_1}) - (x - x_{P_1})(y_{P_2} - y_{P_1}) = 0. \tag{2}$$

For such lines the system 1 can be solved once and for all. The following formula is classical in substance; we include its short proof because everything below builds on it.

Proposition 1 (Point of contact)**.** *Let $P_1(t) = e^{it}$ and $P_2(t) = e^{i\varphi(t)}$ with a C^1 function φ, and assume $P_1(t) \neq P_2(t)$ and $\varphi'(t) \neq -1$. Then the line through $P_1(t)$ and $P_2(t)$ touches the envelope of the family at*

$$Q(t) = \frac{\varphi'(t)\, e^{it} + e^{i\varphi(t)}}{1 + \varphi'(t)},$$

i.e. at the point that divides the chord P_1P_2 in the ratio $1 : \varphi'(t)$.

Proof. With $\theta = \frac{t+\varphi}{2}$ and $\delta = \frac{\varphi - t}{2}$, the chord has midpoint $\cos\delta\, e^{i\theta}$ and the normal form $x\cos\theta + y\sin\theta = \cos\delta$. Differentiating with respect to t, using $\theta' = \frac{1+\varphi'}{2}$ and $\delta' = \frac{\varphi'-1}{2}$, gives $-x\sin\theta + y\cos\theta = \frac{1-\varphi'}{1+\varphi'}\sin\delta$. Hence $Q = e^{i\theta}(\cos\delta + i\frac{1-\varphi'}{1+\varphi'}\sin\delta)$, and inserting $e^{it} = e^{i\theta}e^{-i\delta}$ and $e^{i\varphi} = e^{i\theta}e^{i\delta}$ shows that this equals the stated expression. □

The proof also gives the *pedal curve* of the envelope with respect to the origin, i.e. the locus of the feet of the perpendiculars dropped from the centre onto the tangent lines [6]. Since the perpendicular from the centre bisects every chord, the foot on the line P_1P_2 is simply the midpoint $\frac{1}{2}(P_1 + P_2) = \cos\delta\, e^{i\theta}$. Envelope and pedal curve are easily confused in numerical experiments: intersecting a line with its perpendicular through the origin gives the pedal point, while intersecting it with a neighbouring line of the family gives, in the limit, the point of contact.

The latter observation is also how we draw envelopes numerically. Let P_1, P_2 be the endpoints of the chord at parameter t and Q_1, Q_2 those at $t+\epsilon$. We define $a_{11} = y_{P_1} - y_{P_2}$ and $a_{12} = x_{P_2} - x_{P_1}$ and $a_{21} = y_{Q_1} - y_{Q_2}$ and $a_{22} = x_{Q_2} - x_{Q_1}$. Additionally we define two determinants:

$$det_1 = \begin{vmatrix} x_{P_1} & y_{P_1} \\ x_{P_2} & y_{P_2} \end{vmatrix} \qquad det_2 = \begin{vmatrix} x_{Q_1} & y_{Q_1} \\ x_{Q_2} & y_{Q_2} \end{vmatrix}$$

The intersection (x, y) of the lines P_1P_2 and Q_1Q_2 solves the following system of linear equations:

$$a_{11}x + a_{12}y + det_1 = 0$$
$$a_{21}x + a_{22}y + det_2 = 0$$

By Cramer's rule, with det the coefficient determinant and det_x, det_y the determinants of Cramer's rule, the points of the envelope are obtained as the limits

$$(3)\quad P_{ins}(t) = (x(t), y(t)) = \left(\lim_{\epsilon\to 0}\frac{det_x}{det}, \lim_{\epsilon\to 0}\frac{det_y}{det}\right) = \left(\lim_{\epsilon\to 0}\frac{\begin{vmatrix} a_{12} & det_1 \\ a_{22} & det_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \lim_{\epsilon\to 0}\frac{\begin{vmatrix} det_1 & a_{11} \\ det_2 & a_{21} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}\right)$$

In practice a small fixed ϵ suffices to draw the envelope, see Figure 9 for an example.

3. The doubling and tripling maps

The doubling and tripling maps $x \mapsto 2x \bmod 1$ and $x \mapsto 3x \bmod 1$ on the circle are very sensitive to initial conditions, yet possess dense sets of (repelling) periodic points. Their joint behaviour is the subject of the following heuristic principle going back to Furstenberg [7,8], in the concise form used by Shmerkin [9]:

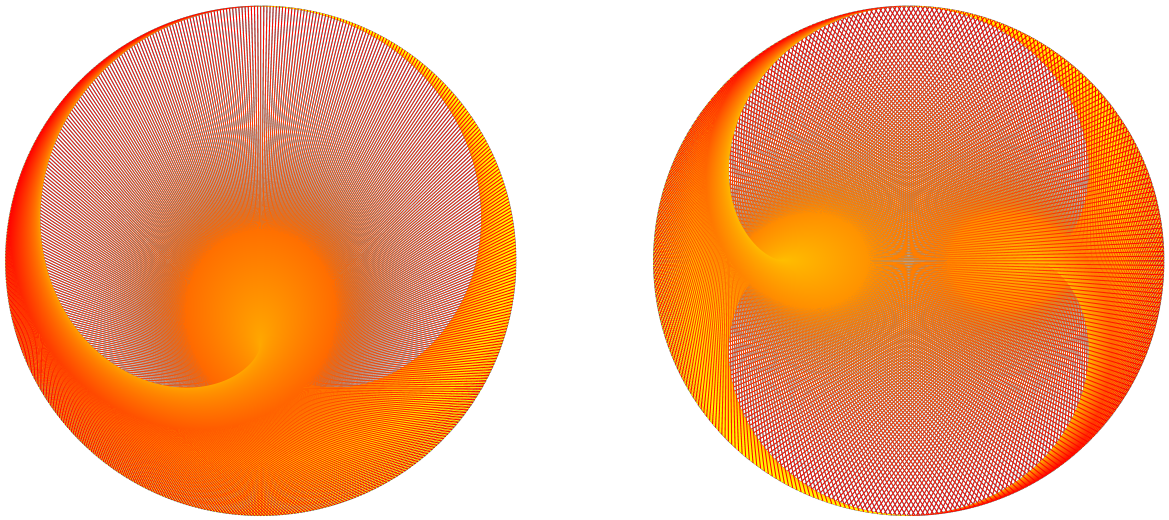
Principle 1 (Furstenberg)**.** ***Expansions in bases 2 and 3 have no common structure.***
More generally, this is expected for bases p and q which are not powers of a common integer or, equivalently, for which $\log p/\log q$ is irrational.

Furstenberg proved that a closed subset of the circle that is invariant under both maps is either finite or the whole circle [7], and he conjectured the measure-theoretic analogue: a Borel probability measure that is invariant and ergodic under both maps is either the Lebesgue measure or supported on a finite set of rational points. Rudolph proved this under the additional assumption of positive entropy [10]; the general case is open, see [11]. A well-known arithmetic manifestation of the principle is the $3x + 1$ or Collatz problem [12], whose dynamics can be described entirely by digit operations in bases 2 and 3:

- if a number ends with digit 1 in base 2, append the digit 1 to its base 3 representation (this computes $3x + 1$);
- delete all consecutive digits 0 at the right end of the base 2 representation (this divides by the largest power of 2).

The standard way of plotting orbits of the doubling or tripling map emphasizes their chaotic behaviour. In this section we draw instead their dreamcatchers in the

standard labelling $f(x) = x$, compute their envelopes and relate them to classical caustics; Section 4 then changes the labelling.



Doubling map $\times 2$: cardioid Tripling map $\times 3$: nephroid

FIGURE 1. Caustics generated by the doubling and the tripling map on the unit circle (rendering: M. Henkel [13])

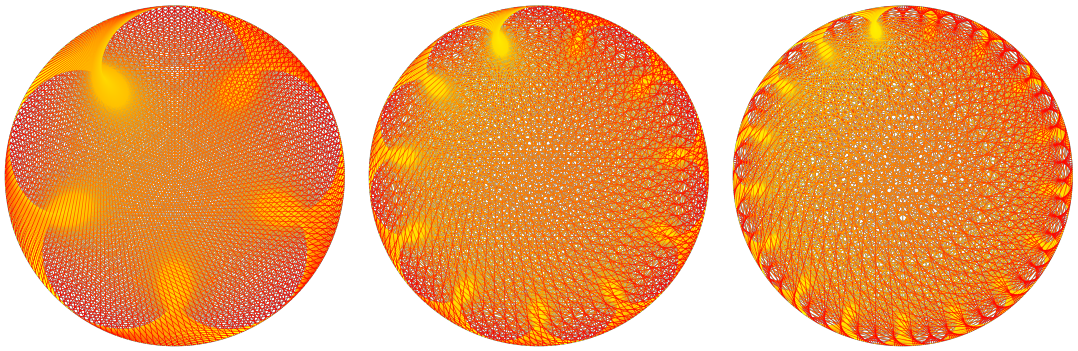
With $X = [0, 1)$ mapped onto the unit circle by $f(x) = x$, i.e. x placed at angle $2\pi x$, Figure 1 (left) shows the *dreamcatcher* of the doubling map $g(x) = 2x \bmod 1$, with a colour gradient indicating the pre-image (red) and the image (yellow). In both panels of Figure 1, the point 0 (identified with 1) sits at the top of the circle and $1/2$ at the bottom. The envelope of the doubling map is a cardioid, which touches the circle at 0 and has its cusp on the diameter through 0 and $1/2$, at distance $1/3$ from the centre on the side of $1/2$ (Lemma 1). The cardioid is also the caustic of a circular mirror for a point light source located on the circle at the position of 0: a ray emitted from 0 that hits the circle at angle b is reflected along the chord from b to $2b$, because the arcs on both sides of the point of reflection are equal. As Figure 1 (right) shows, the envelope of the tripling map is in turn a caustic of the circle, now for parallel rays, i.e. for a light source at infinity.

The Lebesgue measure μ is preserved by the doubling map for an elementary reason, which the mirror symmetry of the cardioid about the line through 0 and $1/2$ illustrates: although $\mu(g(A)) = 2\mu(A)$ for every interval A of length at most $1/2$, the pre-image $g^{-1}(A)$ consists of two disjoint intervals of half the length, so that $\mu(g^{-1}(A)) = \mu(A)$.

If we now represent the dreamcatcher of $g(x) = 3x \bmod 1$ in the same base $f(x) = x$, its envelope is the nephroid, the caustic of the circle for parallel light rays perpendicular to the diameter through 0 and 1/2: a ray along the chord from $-b$ to b is reflected along the chord from b to $3b$.

Here too, the tripling map preserves the Lebesgue measure: for every interval A of length at most 1/3 we have $\mu(g(A)) = 3\mu(A)$, while $g^{-1}(A)$ consists of three disjoint intervals of a third of the length, so that $\mu(g^{-1}(A)) = \mu(A)$. The nephroid has two cusps, which lie on the diameter through $1/4$ and $3/4$ at distance $1/2$ from the centre, and two axes of mirror symmetry, the lines through 0 and $1/2$ and through $1/4$ and $3/4$.

If we now consider the dreamcatchers of maps of the form $x \mapsto 3^n 2^m x \bmod 1$, still in the base $f(x) = x$, we obtain enough insight to state some easily provable lemmas regarding their form.



×6 ×12 ×18

FIGURE 2. Caustics for the ×6, ×12 and ×18 maps (rendering: M. Henkel [13])

Here we see that the envelope of any map of the form $x \mapsto kx \bmod 1$ with $k = 3^n 2^m$ is a single curve with $k - 1$ cusps evenly distributed around the centre. Just like the nephroid of the ×3 map has two axes of symmetry and the cardioid of the ×2 map has one, it has as many symmetry axes as cusps. The following two lemmas are classical [6, 14]; we include short proofs.

Lemma 1. *Let $k \geq 2$ be an integer, for instance $k = 3^n 2^m$. The arithmetic envelope of $x \mapsto kx \bmod 1$ in the base $f(x) = x$ is the epicycloid*

$$z(t) = \frac{k\, e^{it} + e^{ikt}}{k+1}, \qquad t \in [0, 2\pi).$$

It has $k-1$ cusps. They lie on the circle $|z|=\frac{k-1}{k+1}$, at the parameters t for which the chord from t to kt is a diameter. Its symmetry group is the dihedral group D_{k-1}, so it has $k-1$ axes of mirror symmetry and $k-1$ rotational symmetries.

Proof. Proposition 1 with $\varphi(t)=kt$ gives the point of contact $z(t)=\frac{ke^{it}+e^{ikt}}{k+1}$, which is an epicycloid [14, p. 918]. Its derivative $z'(t)=\frac{ik}{k+1}(e^{it}+e^{ikt})$ vanishes if and only if $e^{ikt}=-e^{it}$, i.e. if the chord is a diameter, i.e. if $(k-1)t\equiv\pi \pmod{2\pi}$. This happens for exactly $k-1$ values of $t\in[0,2\pi)$, and then $|z(t)|=\frac{k-1}{k+1}$. The rotation by $\alpha=\frac{2\pi}{k-1}$ maps the chord from t to kt onto the chord from $t+\alpha$ to $kt+\alpha=k(t+\alpha)-(k-1)\alpha\equiv k(t+\alpha)$, which belongs to the family again, and the reflection $t\mapsto -t$ preserves the family as well. Together they generate D_{k-1}. □

Lemma 2. *Let $k=3^n2^m$ and $0\le j<k$. The map $g(x)=kx \bmod 1$ maps the segment $A=[\frac{j}{k},\frac{j+1}{k}]$ onto the entire circle, i.e. onto a set of measure 1. The pre-image $g^{-1}(A)$ consists of k pairwise disjoint segments of length $\frac{1}{k^2}$, one in each segment $[\frac{i}{k},\frac{i+1}{k}]$, so that $\mu(g^{-1}(A))=\mu(A)$ for the Lebesgue measure μ.*

Proof. Directly from $g^{-1}(A)=\bigcup_{i=0}^{k-1}[\frac{i}{k}+\frac{j}{k^2},\frac{i}{k}+\frac{j+1}{k^2}]$. □

Remark 1. The Lebesgue measure is invariant under $x\mapsto kx \bmod 1$, but the converse fails: invariance under a single map of this form does not force a probability measure to be translation-invariant. For example, the point mass at 0, the uniform measure on a periodic orbit (such as $\{1/3,2/3\}$ for $k=2$) and Bernoulli measures on the base-k digits are all invariant under $x\mapsto kx \bmod 1$, but none of them is the Lebesgue measure. The symmetry of an envelope therefore cannot determine the invariant measures. Determining the measures invariant under both $\times2$ and $\times3$ is precisely Furstenberg's conjecture. By contrast, every irrational rotation of the circle is uniquely ergodic, with the Lebesgue measure as its only invariant measure [11].

4. A dyadic coordinate system

So far we have placed $x\in[0,1)$ at the angle $2\pi x$. A first way to grasp the structure linking base 2 and base 3 expansions of integers is to represent the trees generated by the two bases; the binary tree then suggests a different labelling of the circle.

4.1. Two trees and a labelling of the circle.

Definition 2 (n-ary trees over the odd numbers)**.** *Let $\mathbb{N}=\{0,1,2,\dots\}$, so that $2\mathbb{N}+1=\{1,3,5,\dots\}$ is the set of odd positive integers.*

- *The* binary tree *over the odd numbers is the unary algebra $\{2\mathbb{N}+1;\ \cdot2+1,\ \cdot2-1\}$ with root 1.*

- *The* ternary tree *over the odd numbers is the unary algebra* $\{2\mathbb{N}+1;\ \cdot 3,\ \cdot 3+2,\ \cdot 3-2\}$ *with root* 1.

Both are indeed trees: every odd $y > 1$ has exactly one odd parent. In the binary case $y = 2x \pm 1$, where exactly one of $(y \pm 1)/2$ is odd; in the ternary case $y \in \{3x, 3x+2, 3x-2\}$, and the residue of y modulo 3 determines which operation was applied. The branches of each tree, read from the root, are the words in its unary operations. Finite branches correspond to finite words, so there are $\aleph_0$ of them; infinite branches correspond to infinite words, so there are $2^{\aleph_0}$ of them, the cardinality of the continuum. (Informally, one may think of the gap between two consecutive infinite branches, the limit of the spacing between $2^n \cdot x - 1$ and $2^n \cdot x$, as an infinitesimal; we do not use such infinitesimals in the sequel.)

Each branch of each tree captures the full base representation of a number: the operation $\cdot 2 + 1$ appends the digit 1 to the base 2 representation, the operation $\cdot 2 - 1$ appends the digit 0 and then subtracts one, and the operation $\cdot 3$ appends the digit 0 to the base 3 representation. Every number therefore has two branches leading to it, one encoding its base 2 and one its base 3 representation. The binary tree, with the numbers classified by their last digit in base 3, underlies the approach to the Collatz problem in [1]. A three-dimensional rendering, in which the rows of the binary tree are placed on the rings of a cone and the ternary tree is wrapped around it, together with an implementation of the tree structures, is given in [3, Sec. 3–4 and 7].

We now map the binary tree over the odd numbers onto the circle group, which is comparable to a graph embedding. Starting with number 1 in position 0 (equivalently 2π) and number 3 (11 in base 2) in position π, we follow this simple procedure for any number x written in base 2:

- the number x always occupies the same position as $x\underbrace{0\ldots0}_{n}$, i.e. as $2^n x$;
- if the odd number x with $2^n \leq x < 2^{n+1}$ occupies the angle θ, then $2x+1$ occupies the angle $\theta + \pi/2^n$;
- and, for $x > 1$, $2x-1$ occupies the angle $\theta - \pi/2^n$.

Equivalently, x is placed at the angle $2\pi(x/2^{\lfloor \log_2 x \rfloor} - 1)$: the binary digits of x after its leading digit 1, read as a binary fraction, give its position as a fraction of a full turn. For instance, $11 = (1011)_2$ sits at $2\pi \cdot (0.011)_2 = 3\pi/4$. Up to rotation and orientation, this is the distribution of the nodes on each ring of the cone in [3, Def. 6, Fig. 15].

The interest of this approach is primarily to be able to express some diophantine and dynamical arithmetic problems in terms of a well-known topological group, which we hope can be relevant to problems such as the $\times 2$ $\times 3$ conjecture. The embedding of the full binary tree on the circle group can be constructed either as that of $2^{\mathbb{N}}$ branches, each representing a real number, or as the wrapping of the natural number line around

the circle, with the first case being continuous and the second discrete. To better grasp the number distribution we will first describe extensively the discrete case, in which the odd natural numbers are mapped to a particular subgroup of the circle group, that is, the Prüfer 2-group. The transition from the discrete to the continuous case is obtained by taking the closure: the Prüfer 2-group is dense in the circle group. Whenever we consider continuous functions, the full circle group should therefore be assumed to be the base structure, and not any of its subgroups.

4.2. **The Prüfer 2-group.** This coordinate system and its group structure were described in detail in [2]. We recall only what is needed here and refer to [2] for the step-by-step derivation of the group operation, a closed formula, an addition table and a Mathematica implementation.

We label distinguished points of the unit circle in $\mathbb{C}$ counterclockwise with the odd natural numbers, as shown in [2, Fig. 1]: the identity element 1 sits at the angle 0, the number 3 opposite to it at π, and the odd numbers from $2^{n+1}+1$ to $2^{n+2}-1$ fill in the vertices of a regular 2^n-gon (a pair of antipodal points for $5, 7$, a square for 9 to 15, an octagon for 17 to 31, and so forth).

The Prüfer 2-group $\mathbb{Z}(2^\infty) = \mathbb{Z}[1/2]/\mathbb{Z}$ is the group of rationals whose denominator is a power of 2, taken modulo 1; equivalently, it is the group of all 2^n-th roots of unity, $n \in \mathbb{N}$ [15]. Our labelling identifies the odd numbers with its elements. In the notation of Baez [15], with elements $g_1, g_2, \ldots$ satisfying $g_1^2 = e$ and $g_{n+1}^2 = g_n$, and with g_n^k denoting the k-fold group sum of g_n, we have

(4)
$$g_n^k = 2^n + k \qquad \text{with an angle (counterclockwise)} \qquad \pi \cdot k/2^{n-1}, \qquad k \text{ odd},\ 0 < k < 2^n.$$

Adding two elements means adding their angles modulo 2π [15]. Geometrically, this is the *group law on conics* of Lemmermeyer [16, 17]: the sum of two elements is the second intersection of the circle with the parallel through 1 to the chord joining them, see [2, Fig. 3]. For example, $5 + 9 = 11$, $11 + 15 = 5$ and $13 + 19 = 29$. A closed formula for the operation in terms of $\lfloor \log_2(\cdot) \rfloor$ and the 2-adic valuation, an addition table of the first elements and a Mathematica implementation are given in [2, eq. (2), Table 1, Listing 1].[1] The following digit rule gives an equivalent and more transparent description.

[1] We take the opportunity to correct some minor slips in the description in [2]: the labels 5 and 7 correspond to the two cosets $1/4 + \mathbb{Z}$ and $3/4 + \mathbb{Z}$, not to a single coset $1/2 + \mathbb{Z}$, and analogously on the higher levels; the case distinction refers to α/π rather than to α; and the reduction step uses the 2-adic valuation of the numerator. The closed formula and the addition table in [2] are correct.

Proposition 2 (Digit rule)**.** *Write two odd numbers in base 2 as* $x = (1u)_2$ *and* $y = (1v)_2$*, where* u *and* v *are (possibly empty) digit strings. Then* $x + y = (1w)_2$ *in our Prüfer 2-group, where the digit string* w *is obtained as follows:*

(1) align u *and* v *at the left, i.e. read them as binary fractions* $(0.u)_2$ *and* $(0.v)_2$*;*
(2) add these binary fractions digit by digit from right to left, with carries;
(3) discard a carry out of the first digit (this is the reduction modulo 1*);*
(4) delete all trailing digits 0*.*

Proof. By equation 4 the odd number $(1u)_2$ is located at the angle $2\pi \cdot (0.u)_2$. The group operation adds angles modulo 2π, so it adds the binary fractions $(0.u)_2$ and $(0.v)_2$ modulo 1. Deleting trailing zeros and prepending the leading digit 1 inverts this correspondence. □

For example, $0.101 + 0.0011 = 0.1101$ gives $13 + 19 = (11101)_2 = 29$. Carries cannot be ignored, not even when the two numbers have different lengths: $5+11$ corresponds to $0.01 + 0.011 = 0.101$, hence $5 + 11 = 13$, whereas a carry-free digit-wise rule (exclusive or) would give 9. If both numbers have the same length, the rule amounts to ordinary binary addition of x and y, keeping the number of digits of x, setting the first digit to 1 and deleting trailing zeros; for instance $13 + 15 = (11100)_2 \mapsto (1100)_2 \mapsto (11)_2 = 3$.

4.3. **Drawing envelopes within the Prüfer 2-group.** Let us begin with number 9. The first line extends from 9 to the number 27, since $9 \times 3 = 27$. For simplicity, we use 11 as the next number to multiply by 3, but the envelope is generated by applying the action to all odd numbers. Multiplied by 3, the result and therefore the endpoint of the second line is 33. The two lines are then intersected. If this is repeated, the intersection points outline a shape within the unit circle, see Figure 3. The chords from x to $3x$ with $2^n \le x < 2^{n+1}$ belong to one and the same family of lines for every n (Lemma 3), so higher levels only sample this family more densely. The complete dreamcatcher of $\times 3$ in this labelling, with the vertices coloured by their residue class modulo 3, is shown in [2, Fig. 2].

That these chords leave a convex region in the centre free was first observed by Henkel, who called it the *convexoid* and suggested that it is a truncated caustic [3, Sec. 7.1, Figs. 24–26]. Theorem 1 in Section 5 makes this precise: the region is bounded by arcs of two epicycloids and by a segment of the diameter joining the labels 1 and 3, which accounts for the truncation.

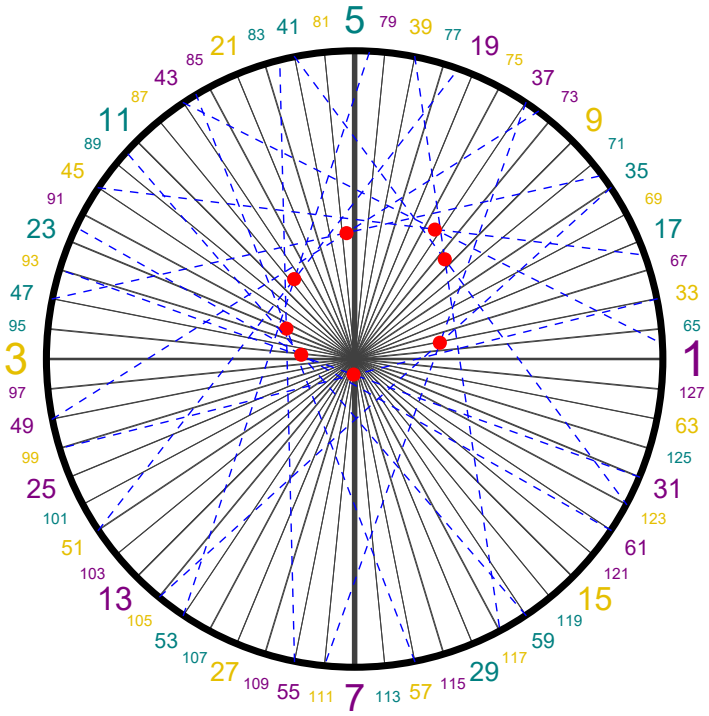


FIGURE 3. Construction of the envelope of the action $\cdot 3$ in the labelled Prüfer 2-group: labels up to 2^7, with the shape of the $\times 3$ dreamcatcher already outlined; cf. [3, Fig. 26] and [2, Fig. 2]

5. The convexoid of multiplication by 3

Figure 3 and Henkel's renderings [3, Figs. 24–26] show that the chords from x to $3x$ in the dyadic labelling leave a region in the centre of the circle free. In this section we determine this region exactly. Angles are measured counterclockwise, and the odd number x with binary mantissa $m = x/2^{\lfloor \log_2 x \rfloor} \in [1, 2)$ sits at the angle $t = 2\pi(m-1)$.

5.1. Multiplication by 3 as a circle map.

Lemma 3. *If x sits at the angle t, then $3x$ sits at the angle $\Phi(t)$ modulo 2π, where*

$$\Phi(t) = \begin{cases} \frac{3}{2}t + \pi, & 0 \le t < \frac{2\pi}{3}, \\ \frac{3}{4}t - \frac{\pi}{2}, & \frac{2\pi}{3} \le t < 2\pi. \end{cases}$$

In particular, the chords from x to $3x$ with $2^n \le x < 2^{n+1}$ are the members L_t, $t \in 2\pi \cdot 2^{-n}\mathbb{Z} \cap [0, 2\pi)$, of the single family of chords L_t from e^{it} to $e^{i\Phi(t)}$, $t \in [0, 2\pi)$.

Proof. The mantissa of $3x$ is $\frac{3}{2}m$ if $3m < 4$, i.e. if $t < \frac{2\pi}{3}$, and $\frac{3}{4}m$ otherwise. Inserting $m = 1 + \frac{t}{2\pi}$ gives the two branches. □

Proposition 3. *Φ is an orientation-preserving piecewise linear homeomorphism of the circle with slopes $\frac{3}{2}$ and $\frac{3}{4}$ and break points at $t = 0$ and $t = \frac{2\pi}{3}$. The map $h(t) = 2\pi \log_2(1 + \frac{t}{2\pi})$ conjugates it to a rotation:*

$$h(\Phi(t)) \equiv h(t) + 2\pi \log_2 3 \pmod{2\pi}.$$

Consequently, Φ has the irrational rotation number $\log_2 3 \bmod 1$, it has no periodic points, and all its orbits are dense.

Proof. Both branches agree modulo 2π at the break points, and both slopes are positive. In terms of the mantissa, Φ is the map $m \mapsto 3m/2^j$ with $j \in \{1, 2\}$ chosen such that $3m/2^j \in [1, 2)$, and h corresponds to $m \mapsto 2\pi \log_2 m$. Hence $h \circ \Phi$ corresponds to $2\pi(\log_2 m + \log_2 3 - j)$. Since $\log_2 3$ is irrational, every orbit of the rotation is dense [18]. □

Proposition 3 is the familiar mechanism behind Benford's law for the leading digits of the powers of 3 [19,20]. In our setting it is the dyadic incarnation of Furstenberg's principle, the irrationality of $\log 3/\log 2$. If one places x at the logarithmic angle $2\pi \log_2 x$ instead, multiplication by 3 becomes the rotation itself, all chords from x to $3x$ have the same length, and they leave free exactly the open disk of radius $|\cos(\pi \log_2 3)| \approx 0.2638$ (Lemma 4 below and Figure 4, right). The dyadic labelling distorts this disk into the convexoid.

5.2. **Free regions of chord families.** For $z, w \in \mathbb{C}$ we write $\langle z, w\rangle = \mathrm{Re}(z\overline{w})$ for the Euclidean inner product, and $\mathbb{D}$ for the open unit disk.

Lemma 4 (Free regions are convex)**.** *Let $\varphi\colon \mathbb{R} \to \mathbb{R}$ be continuous with $\varphi(t + 2\pi) = \varphi(t) + 2\pi$ and $\pi \le \varphi(t) - t < 2\pi$ for all t. Let L_t be the chord from e^{it} to $e^{i\varphi(t)}$, and put $\theta(t) = \frac{t+\varphi(t)}{2}$ and $\delta(t) = \frac{\varphi(t)-t}{2}$. Then a point $p \in \mathbb{D}$ lies on none of the chords L_t if and only if*

$$\langle p, e^{i\theta(t)}\rangle > \cos\delta(t) \qquad \text{for all } t \in \mathbb{R}. \tag{5}$$

In particular, the free region $F = \{p \in \mathbb{D} : p \notin L_t \text{ for all } t\}$ is open and convex, and it is contained in the disk of radius $\max_t |\cos\delta(t)|$.

Proof. The line through e^{it} and $e^{i\varphi(t)}$ is $\{z : \langle z, e^{i\theta(t)}\rangle = \cos\delta(t)\}$, and it meets the closed unit disk exactly in L_t. Hence $p \in \mathbb{D}$ lies on L_t if and only if $g_p(t) = 0$, where $g_p(t) = \langle p, e^{i\theta(t)}\rangle - \cos\delta(t)$. The function g_p is continuous and 2π-periodic. Since $\theta(t+2\pi) = \theta(t)+2\pi$, there is a t_1 with $p = |p|e^{i\theta(t_1)}$, and then $g_p(t_1) = |p| - \cos\delta(t_1) \ge 0$, because $\frac{\pi}{2} \le \delta < \pi$. By the intermediate value theorem, g_p has no zero if and only

if it is positive everywhere, which is (5). The minimum of g_p over a period depends continuously on p, so F is open, and as an intersection of half-planes it is convex. Finally, if $0 \neq p \in F$, choose t_2 with $-p = |p|e^{i\theta(t_2)}$; then (5) gives $|p| < -\cos\delta(t_2)$. □

For the logarithmic labelling, $\varphi(t) = t + 2\pi(\log_2 3 - 1)$, and δ is constant with $\cos\delta = -|\cos(\pi \log_2 3)|$; condition (5) then describes the disk mentioned above.

5.3. **The convexoid.**

Theorem 1 (Henkel's convexoid)**.** *Let F be the set of points of $\mathbb{D}$ that lie on none of the chords L_t, $t \in [0, 2\pi)$, of Lemma 3, and put*

$$Q_r(t) = \frac{3e^{it} - 2e^{3it/2}}{5}, \qquad Q_b(t) = \frac{3e^{it} - 4i\, e^{3it/4}}{7}.$$

Then F is an open convex set whose boundary consists of

(1) the segment $[-\frac{1}{7}, \frac{1}{5}]$ of the diameter through the labels 1 and 3,
(2) the arc $Q_r([0, t_r])$, which starts with a cusp at $\frac{1}{5}$, and
(3) the arc $Q_b([t_b, 2\pi])$, which ends with a cusp at $-\frac{1}{7}$.

The arcs meet at the corner $C = Q_r(t_r) = Q_b(t_b) \approx 0.21545 + 0.42511i$, where $t_r \approx 1.82917$ and $t_b \approx 2.46469$. Here Q_r is an epicycloid with fixed circle of radius $\frac{1}{5}$ and rolling circle of radius $\frac{2}{5}$, and Q_b an epicycloid with fixed circle of radius $\frac{1}{7}$ and rolling circle of radius $\frac{3}{7}$. The area of F equals

$$\frac{3}{10}(t_r - 2\sin\frac{t_r}{2}) + \frac{3}{14}(2\pi - t_b - 4\cos\frac{t_b}{4}) \approx 0.19210.$$

Python programs that reproduce the corner C, the area, the computer-assisted step of the following proof and Figure 4, and that compare F with the heart-shaped pedal curve of Section 6, are available in [21].

Proof. Step 1: convexity. We use the continuous lift $\tilde\Phi(t) = \frac{3}{2}t + \pi$ on $[0, \frac{2\pi}{3}]$ and $\tilde\Phi(t) = \frac{3}{4}t + \frac{3\pi}{2}$ on $[\frac{2\pi}{3}, 2\pi]$, extended by $\tilde\Phi(t + 2\pi) = \tilde\Phi(t) + 2\pi$. Then $\delta(t) = \frac{t}{4} + \frac{\pi}{2}$ on the first and $\delta(t) = \frac{3\pi}{4} - \frac{t}{8}$ on the second interval, so $\frac{\pi}{2} \le \delta \le \frac{2\pi}{3}$ and Lemma 4 applies. Hence F is open and convex and lies in the disk of radius $\frac{1}{2}$. It is not empty: for $p = i\varepsilon$ with $0 < \varepsilon < \frac{1}{5}$ one has $g_p(t) = \varepsilon\cos\frac{5t}{4} + \sin\frac{t}{4}$ on $[0, \frac{2\pi}{3}]$ and, with $s = 2\pi - t$, $g_p(t) = \varepsilon\cos\frac{7s}{8} + \sin\frac{s}{8}$ on $[\frac{2\pi}{3}, 2\pi]$, and both expressions are positive (where the cosine is negative, the sine term exceeds $\sin\frac{\pi}{14} > \frac{1}{5}$). Thus ∂F is a convex Jordan curve.

Step 2: boundary points are points of contact. On the open intervals $(0, \frac{2\pi}{3})$ and $(\frac{2\pi}{3}, 2\pi)$ the chords have $\tilde\Phi' = \frac{3}{2}$ resp. $\frac{3}{4}$, and Proposition 1 gives their points of contact

$Q(t) = Q_r(t)$ resp. $Q(t) = Q_b(t)$. The derivative $g_p'(t)$ is affine in p with gradient $\theta'(t)\, ie^{i\theta(t)}$ and vanishes at $p = Q(t)$, so

$$g_p'(t) = \theta'(t)\, \langle p - Q(t),\, ie^{i\theta(t)} \rangle, \qquad \theta'(t) > 0.$$

Let $p \in \partial F$. Then $g_p \geq 0$, and since $p \notin F$, g_p attains the value 0 at some t_0. If t_0 lies in one of the open intervals, then $g_p(t_0) = 0$ and $g_p'(t_0) = 0$; since $p - Q(t_0)$ is parallel to the chord, i.e. to $ie^{i\theta(t_0)}$, this forces $p = Q(t_0)$. If $t_0 = \frac{2\pi}{3}$, a minimum requires $\langle p - Q_r(\frac{2\pi}{3}), \tau \rangle \leq 0 \leq \langle p - Q_b(\frac{2\pi}{3}), \tau \rangle$ with $\tau = ie^{i\theta(2\pi/3)} = e^{-i\pi/6}$. This is impossible, because $Q_r(\frac{2\pi}{3}) = \frac{1+3\sqrt{3}\,i}{10}$, $Q_b(\frac{2\pi}{3}) = \frac{5+3\sqrt{3}\,i}{14}$ and $\langle Q_b(\frac{2\pi}{3}) - Q_r(\frac{2\pi}{3}), \tau \rangle = \frac{6\sqrt{3}}{35} > 0$. If $t_0 = 0$, then $g_p(0) = \operatorname{Im} p = 0$, and the one-sided conditions $g_p'(0^+) \geq 0 \geq g_p'(2\pi^-)$ read $\frac{1}{5} - p \geq 0$ and $-\frac{1}{7} - p \leq 0$. Hence

$$\partial F \subset [-\frac{1}{7}, \frac{1}{5}] \cup \Gamma_r \cup \Gamma_b, \qquad \Gamma_r = Q_r([0, \frac{2\pi}{3}]), \quad \Gamma_b = Q_b([\frac{2\pi}{3}, 2\pi]).$$

Step 3: the two arcs. A short computation gives

$$\operatorname{Im} Q_r(t) = \frac{2}{5} \sin\frac{t}{2}\,(1 - \cos\frac{t}{2})(1 + 4\cos\frac{t}{2}),$$
$$\operatorname{Im} Q_b(t) = \frac{4}{7} \sin w\,(1 - \cos w)(6\cos^2 w + 2\cos w - 1),$$

with $w = \frac{\pi}{2} - \frac{t}{4}$. Both are positive for $0 < t \leq \frac{2\pi}{3}$ resp. $\frac{2\pi}{3} \leq t < 2\pi$, so the arcs meet the real axis only in their endpoints $Q_r(0) = \frac{1}{5}$ and $Q_b(2\pi) = -\frac{1}{7}$. Since $Q'(t) = \frac{2\lambda\cos\delta(t)}{1+\lambda}\, ie^{i\theta(t)}$ with $\lambda = \tilde{\Phi}'$, the arcs are regular except at these two cusps, and along Γ_r the tangent turns by $\theta(\frac{2\pi}{3}) - \theta(0) = \frac{5\pi}{6} < \pi$, so Γ_r is a simple arc. Writing $u = e^{it/2}$ on Γ_r and $w = e^{it/4}$ on Γ_b, the conditions for a self-intersection of Γ_b and for an intersection of Γ_r and Γ_b become polynomial equations on the unit circle. A resultant computation (Appendix C) shows that Γ_b is simple and that $\Gamma_r \cap \Gamma_b = \{C\}$ with the stated parameters.

Step 4: conclusion. By Steps 2 and 3, the Jordan curve ∂F is contained in the plane graph with vertices $-\frac{1}{7}, \frac{1}{5}, C, Q_r(\frac{2\pi}{3}), Q_b(\frac{2\pi}{3})$ and edges $[-\frac{1}{7}, \frac{1}{5}]$, $Q_r([0, t_r])$, $Q_r([t_r, \frac{2\pi}{3}])$, $Q_b([\frac{2\pi}{3}, t_b])$, $Q_b([t_b, 2\pi])$. The vertices $Q_r(\frac{2\pi}{3})$ and $Q_b(\frac{2\pi}{3})$ have degree one, so the only cycle of this graph is formed by the segment and the arcs $Q_r([0, t_r])$ and $Q_b([t_b, 2\pi])$. A Jordan curve contained in a finite plane graph is a cycle of the graph, so ∂F is this cycle. Moreover, $Q_r(t) = (R + r)e^{it} - re^{i(R+r)t/r}$ with $R = \frac{1}{5}$, $r = \frac{2}{5}$, and with $\sigma = \frac{3}{4}t - \frac{3\pi}{2}$ one checks $Q_b(t) = -((R + r)e^{i\sigma} - re^{i(R+r)\sigma/r})$ with $R = \frac{1}{7}$, $r = \frac{3}{7}$; these are the standard parametrizations of epicycloids [6]. Finally, on the segment $y\,dx = x\,dy = 0$, while $x\,dy - y\,dx = \frac{3}{5}(1 - \cos\frac{t}{2})\,dt$ on Q_r and $\frac{3}{7}(1 - \sin\frac{t}{4})\,dt$ on Q_b, and Green's formula gives the area. □

Corollary 1. *A point of $\mathbb{D}$ lies in the closure of the union of all chords from x to $3x$, x odd, if and only if it does not lie in F.*

Proof. By Lemma 3, these chords are the L_t with $t/2\pi$ a dyadic rational, which are dense in $[0, 2\pi)$. The chord L_t depends continuously on $t \in \mathbb{R}/2\pi\mathbb{Z}$, also at the break point $\frac{2\pi}{3}$, where both branches give the chord to the label 1, and at $t = 0$, where L_t tends to the diameter L_0. Hence the closure of the union of the integer chords is the union of all L_t, which is compact and has complement F in $\mathbb{D}$. □

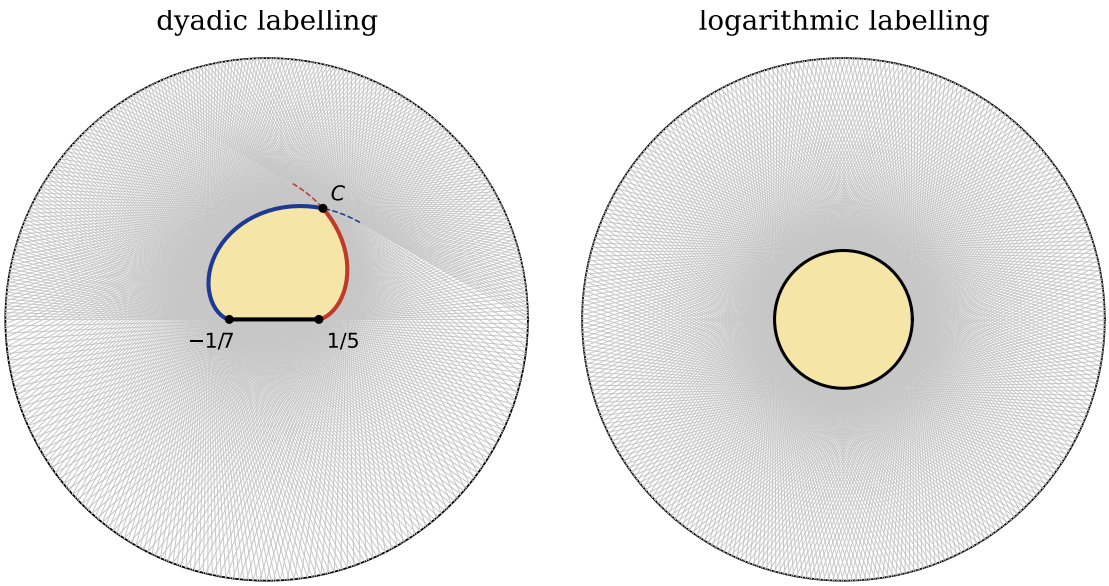


FIGURE 4. Left: the chord family of Lemma 3 and the convexoid F of Theorem 1, bounded by the segment $[-\frac{1}{7}, \frac{1}{5}]$ and arcs of Q_r (red) and Q_b (blue); the dashed continuations of the arcs lie outside F. Right: in the logarithmic labelling the free region is the disk of radius $|\cos(\pi \log_2 3)|$.

Remark 2. The numbers 5 and 7 in Theorem 1 are the same as in the Chebyshev polynomials T_5 and T_7 of Section 6: for a chord family with slope λ and $\kappa = \frac{\lambda-1}{\lambda+1}$, the envelope has its cusps on the circle of radius $|\kappa|$, and the pedal curve is the rose $\rho = \cos(\kappa\vartheta + \beta)$ for a suitable phase β; the slopes $\lambda = \frac{3}{2}$ and $\lambda = \frac{3}{4}$ give $\kappa = \frac{1}{5}$ and $\kappa = -\frac{1}{7}$. Every point of the pedal curve is the midpoint of a chord and therefore lies outside F. Numerically, F lies in the region enclosed by the heart-shaped pedal curve and touches it only at the origin [21].

6. The heart-shaped pedal curve of $\times 3$

Besides the convexoid, the chord family of Lemma 3 has a second natural companion curve: its pedal curve, which is heart-shaped and encloses the convexoid. It is created as follows. We use the unit circle with the vertices labelled as in Section 4.2, see also [2, Fig. 1]. The labelling places the odd number x at the angle $2\pi(g(x)-1)$, where $g(x) = {}^{x}/_{2^{\lfloor \log_2(x) \rfloor}} \in [1,2)$ is the binary mantissa of x. We connect a vertex $P_1 = (\cos t, \sin t)$ labelled by x with the vertex P_2 labelled by $3x$. The line connecting P_1 and P_2 has the form $y = m(t)x + n(t)$. These lines are the tangents of the arithmetic envelope of $\times 3$ in this labelling, which consists of two arcs of epicycloids. The heartshape is not this envelope itself but its *pedal curve* with respect to the origin, i.e. the locus of the feet of the perpendiculars dropped from the centre of the circle onto the tangents (Section 2). The perpendicular to the tangent through the origin has the slope ${}^{-1}/_{m(t)}$, so it is the line $y = {}^{-1}/_{m(t)} \cdot x$. Intersecting the tangent with this perpendicular gives a point of the pedal curve:

$$\begin{aligned} &\text{tangent:} && y = m(t)x + n(t) \\ &\text{normal:} && y = -\frac{1}{m(t)}x \\ &\text{intersection:} && P_{ins}(t) = (x(t), y(t)) = \left(-\frac{m(t) \cdot n(t)}{m(t)^2+1}, \frac{n(t)}{m(t)^2+1} \right) \end{aligned} \tag{6}$$

We obtain the intersection of the tangent passing through P_1 and P_2 with the y-axis by inserting P_1 into the linear equation of the tangent $y = m(t)x + n(t)$ which results in $n(t) = y - m(t)x = \sin(t) - m(t)\cos(t)$.

Now we can formulate a point of the heartshape as the intersection of the tangent with the perpendicular through the origin, see equation 7:

$$P_{ins}(t) = (x(t), y(t)) = \left(\frac{m(t)^2 \cos(t) - m(t)\sin(t)}{m(t)^2+1}, \frac{\sin(t) - m(t)\cos(t)}{m(t)^2+1} \right) \tag{7}$$

From 6 we know that $m(t) = {}^{-x}/_{y}$, which we insert into $n(t) = \sin(t) - m(t)\cos(t)$ and obtain $n(t) = \sin(t) + {}^{x}/_{y}\cos(t)$ respectively $y \cdot n(t) = y\sin(t) + x\cos(t)$. Furthermore, equation 6 provides the equality $y \cdot n(t) = y\,(y - m(t)x) = y^2 - m(t)xy = y^2 - {}^{-x}/_{y} \cdot xy$ leading to the following equation, which we call the normal equation (through P_1); it describes the circle with diameter OP_1, on which the foot of the perpendicular lies by Thales' theorem:

$$x^2 + y^2 = y\sin t + x\cos t \tag{8}$$

In this section we measure the angle t clockwise, which amounts to reflecting the labelling of [2, Fig. 1] in the horizontal axis. This does not affect the resulting polynomial equations, which contain only even powers of y. Consider the labels $2^6 \le x < 2^7$. If $x \le 85$, i.e. $3x < 2^8$, and the vertex labelled x has the angle t, then the vertex labelled $3x$ has the angle $3t/2$. If $85 < x$, it has the angle $3t/4 + \pi$ (modulo 2π). The threshold $85 \approx \frac{4}{3} \cdot 2^6$ corresponds to $t = 4\pi/3$, and the same two formulas hold on every level $2^n \le x < 2^{n+1}$ with the threshold $\frac{4}{3} \cdot 2^n$, because the binary mantissa of $3x$ depends only on the mantissa of x. For this reason the slope $m(t)$ of the line connecting both vertices P_1 and P_2 is defined as follows:

$$m(t) = \begin{cases} \dfrac{\sin\frac{3t}{2} - \sin t}{\cos\frac{3t}{2} - \cos t}, & \frac{4\pi}{3} < t < 2\pi \qquad \text{red curve piece} \\[2ex] \dfrac{\sin\frac{3t}{4} + \sin t}{\cos\frac{3t}{4} + \cos t}, & 0 < t < \frac{4\pi}{3} \qquad \text{blue curve piece} \end{cases} \tag{9}$$

The heartshape has two curve pieces, which we color red and blue, see Figure 5. Note that we simplified the sin and cos terms in the slope of the blue curve piece by setting $\sin(3t/4 + \pi) = -\sin(3t/4)$ and $\cos(3t/4 + \pi) = -\cos(3t/4)$.

6.1. **Polynomial equation of the red curve piece.** Starting point for determining the polynomial equation of the red curve piece is the system of normal equations as defined by equation 8, which leads to the equation for zeros:

$$\begin{aligned} x^2 + y^2 &= y\sin t + x\cos t \\ x^2 + y^2 &= y\sin\tfrac{3t}{2} + x\cos\tfrac{3t}{2} \\ 0 &= y\sin\tfrac{3t}{2} + x\cos\tfrac{3t}{2} - y\sin(t) - x\cos(t) \end{aligned} \tag{10}$$

We isolate a linear equation in x and y for each value t and get the perpendicular through the origin, $y = -\frac{1}{m(t)}x$ with

$$-\frac{1}{m(t)} = \frac{\cos t - \cos\frac{3t}{2}}{\sin\frac{3t}{2} - \sin t} = \frac{2\sin\frac{5t}{4}\sin\frac{t}{4}}{2\cos\frac{5t}{4}\sin\frac{t}{4}} = \tan\frac{5t}{4}.$$

Recall that the domain of our red curve is $4\pi/3 < t < 2\pi$. A division by zero cannot occur, since the equality $\sin\frac{3t}{2} - \sin t = 0$ never happens in this domain. Based on this ratio, we define $x(t)$ and $y(t)$ in terms of the unknown magnitude $r(t)$ as $x(t) = r\cos 5t/4$ and $y(t) = r\sin 5t/4$. From the first equation in the system of normal equations 10 we obtain $r^2 = r\sin 5t/4\sin t + r\cos 5t/4\cos t$. Dividing r out leads to $r = \cos(5t/4 - t) = \cos t/4$ and therefore to $x = \cos 5t/4\cos t/4$ and $x^2 + y^2 = \cos^2 t/4$.

The fifth Chebyshev polynomial of the first kind is $T_5(z) = 16z^5 - 20z^3 + 5z$. Inserting $z = \cos t/4$ yields:

$$\cos\frac{5t}{4} = T_5\left(\cos\frac{t}{4}\right) = 16\cos^5\frac{t}{4} - 20\cos^3\frac{t}{4} + 5\cos\frac{t}{4}$$

Multiplying both sides of this equation with $\cos t/4$ leads to our searched polynomial:

$$x = 16\cos^6\frac{t}{4} - 20\cos^4\frac{t}{4} + 5\cos^2\frac{t}{4} = 16\left(x^2+y^2\right)^3 - 20\left(x^2+y^2\right)^2 + 5\left(x^2+y^2\right)$$

In other words, with $\rho^2 = x^2 + y^2$ the red piece lies on the curve $x = \rho T_5(\rho)$, which in polar coordinates (ρ, ϑ) is the rose $\rho = \cos\frac{\vartheta}{5}$, since $\vartheta = \frac{5t}{4}$ and $\rho = \cos\frac{t}{4}$.

We obtain the same polynomial using a computational approach, which we will apply to other curves later. For this we solve the system of equations 10 for x and y. Since the perpendicular from the centre bisects every chord, the solution is simply the midpoint of the chord P_1P_2:

$$\begin{aligned} x &= \tfrac{1}{2}\left(\cos t + \cos\tfrac{3t}{2}\right) = 2\cos^3\tfrac{t}{2} + \cos^2\tfrac{t}{2} - \tfrac{3}{2}\cos\tfrac{t}{2} - \tfrac{1}{2} \\ y &= \tfrac{1}{2}\left(\sin t + \sin\tfrac{3t}{2}\right) = 2\sin\tfrac{t}{2}\cos^2\tfrac{t}{2} + \sin\tfrac{t}{2}\cos\tfrac{t}{2} - \tfrac{1}{2}\sin\tfrac{t}{2} \end{aligned} \tag{11}$$

We obtain the above-shown new system of equations 11 by calling in Maxima CAS:

```
solve([x^2+y^2-y*sin(t)-x*cos(t), x^2+y^2-y*sin(3*t/2)-x*cos(3*t/2)], [x,y]);
```

The resulting equation for x and y has been simplified[2]. To finally obtain the polynomial equation we substitute $\sin\frac{t}{2} = s$ and $\cos\frac{t}{2} = c$ and apply Buchberger's algorithm in Maxima CAS using `load(grobner)` and

```
poly_buchberger([s^2+c^2-1, x-(2*c^3+c^2-3*c/2-1/2), y-(2*s*c^2+s*c-s/2)], [c,s,x
    ,y]);
```

which yields the polynomial equation $0 = 16y^6 + 48x^2y^4 - 20y^4 + 48x^4y^2 - 40x^2y^2 + 5y^2 + 16x^6 - 20x^4 + 5x^2 - x$ for the red curve piece. The generator $s^2 + c^2 - 1$ encodes the relation $\sin^2 t/2 + \cos^2 t/2 = 1$.

[2]Simplifications can be achieved tool-based, e.g. in Mathematica using `FullSimplify[...]`, in Maple using `convert(..., exp)` and then `simplify(...)`. To split up sums and integer multiples that appear in arguments of trigonometric functions, we use in Mathematica `TrigExpand[...]`.

6.2. **Polynomial equation of the blue curve piece.** Starting point for determining the polynomial equation of the blue curve piece is the system of normal equations (see equation 8), which leads to the equation for zeros:

$$\begin{aligned} x^2+y^2 &= y\sin t + x\cos t \\ x^2+y^2 &= -y\sin\tfrac{3t}{4} - x\cos\tfrac{3t}{4} \\ 0 &= y\sin\tfrac{3t}{4} + x\cos\tfrac{3t}{4} + y\sin(t) + x\cos(t) \end{aligned} \tag{12}$$

Using the computational approach we solve the system of equations 12 and obtain:

$$\begin{aligned} x &= \tfrac{1}{2}\left(\cos t - \cos\tfrac{3t}{4}\right) = 4\cos^4\tfrac{t}{4} - 2\cos^3\tfrac{t}{4} - 4\cos^2\tfrac{t}{4} + \tfrac{3}{2}\cos\tfrac{t}{4} + \tfrac{1}{2} \\ y &= \tfrac{1}{2}\left(\sin t - \sin\tfrac{3t}{4}\right) = 4\sin\tfrac{t}{4}\cos^3\tfrac{t}{4} - 2\sin\tfrac{t}{4}\cos^2\tfrac{t}{4} - 2\sin\tfrac{t}{4}\cos\tfrac{t}{4} + \tfrac{1}{2}\sin\tfrac{t}{4} \end{aligned} \tag{13}$$

We substitute $\sin\frac{t}{4} = s$ and $\cos\frac{t}{4} = c$ and by using Buchberger's algorithm we again obtain a polynomial equation:

```
poly_buchberger([s^2+c^2-1, x-(4*c^4-2*c^3-4*c^2+3*c/2+1/2), y-(4*s*c^3-2*s*c
    ^2-2*s*c+s/2)], [c,s,x,y]);
```

Here the generator s^2+c^2-1 encodes $\sin^2 t/4 + \cos^2 t/4 = 1$. The resulting polynomial equation is:

$$\begin{aligned} 0 = &-64y^8 - 256x^2y^6 + 112y^6 - 384x^4y^4 + 336x^2y^4 - 56y^4 - 256x^6y^2 \\ &+336x^4y^2 - 112x^2y^2 + 7y^2 - 64x^8 + 112x^6 - 56x^4 + 7x^2 + x \end{aligned}$$

Analogously to the red piece, this means $x = \rho\, T_7(\rho)$ with $\rho^2 = x^2+y^2$ and $T_7(z) = 64z^7 - 112z^5 + 56z^3 - 7z$: here the chord midpoint has the polar angle $\vartheta = \frac{7t}{8} + \frac{\pi}{2}$ and the radius $\rho = \cos\delta$ with $\delta = \frac{\pi}{2} - \frac{t}{8}$, so that $7\delta \equiv -\vartheta \pmod{2\pi}$ and hence $\cos\vartheta = T_7(\rho)$.

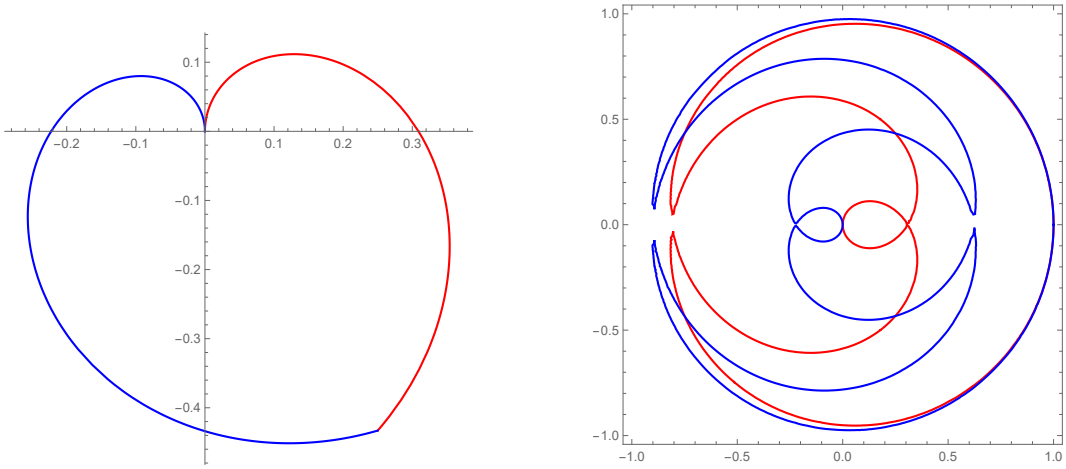


FIGURE 5. Heartshape plotted using the parametric form (left) and the polynomial equations (right)

Appendix A discusses the tropicalization of both polynomials.

7. Further examples

7.1. **Implicit equations of epicycloids.** For the map $x \mapsto ax$ with an integer $a \geq 2$ in the standard labelling we connect $P_1(t) = (\cos t, \sin t)$ with $P_2(t) = (\cos at, \sin at)$, and Proposition 1 gives the epicycloid

$$x(t) = \frac{a \cdot \cos t + \cos(at)}{1+a}$$
$$y(t) = \frac{a \cdot \sin t + \sin(at)}{1+a} \tag{14}$$

The curves defined by the parametric form 14 represent so-called epicycloids, see [14, p. 918]. For each fixed a one can eliminate t by writing $\cos at$ and $\sin at$ as polynomials in $\cos t$ and $\sin t$ and applying Buchberger's algorithm, as in Section 6. An implicit equation that is uniform in a is obtained more elegantly with resultants. We use Hoon Hong's implicitization approach [22] that is based on the Sylvester matrix. Let $F, G \in \mathbb{C}[x, y, z]$ be two functions defined by $F = z^a + az - (a+1)(x+iy)$ and $G = -(a+1)(x-iy)z^a + az^{a-1} + 1$. Then the cartesian form (the polynomial) providing an implicitization of the curve defined by equations 14 is the resultant of F and G with

respect to z, that is the determinant of the Sylvester matrix as given below, where $a_0 = -(a+1)(x+iy)$ and $\overline{a_0} = -(a+1)(x-iy)$ (the upper bar stands for complex conjugation):

$$\text{(15)} \qquad \operatorname{res}_z(F,G) = \begin{vmatrix} 1 & 0 & \cdots & 0 & a & a_0 & & & & \\ & 1 & 0 & \cdots & 0 & a & a_0 & & & \\ & & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & & \\ & & & 1 & 0 & \cdots & 0 & a & a_0 & \\ & & & & 1 & 0 & \cdots & 0 & a & a_0 \\ \overline{a_0} & a & 0 & \cdots & 0 & 1 & & & & \\ & \overline{a_0} & a & 0 & \cdots & 0 & 1 & & & \\ & & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & & \\ & & & \overline{a_0} & a & 0 & \cdots & 0 & 1 & \\ & & & & \overline{a_0} & a & 0 & \cdots & 0 & 1 \end{vmatrix}$$

This resultant can be computed very efficiently, see Listing 5 in Appendix C. For example, for $a = 2$ one obtains the cardioid $27(x^2+y^2)^2 - 18(x^2+y^2) - 8x - 1 = 0$ and for $a = 3$ the nephroid $\left(4(x^2+y^2)-1\right)^3 = 27x^2$.

7.2. **A non-linear map.** We finally consider a non-linear map in the standard labelling. We choose $f(x) = x^3$ that connects a vertex $P_1 = (\cos t, \sin t)$ labeled by x with a vertex P_2 labeled by x^3. Both vertices P_1 and P_2 lie on the unit circle and have the following coordinates:

$$\text{(16)} \qquad P_1(t) = (\cos t, \sin t) \qquad P_2(t) = \left(\cos \frac{t^3}{4\pi^2}, \sin \frac{t^3}{4\pi^2}\right)$$

Note that a vertex labeled with x has an angle $t = 2\pi \cdot x$. A vertex labeled with x^3 has the angle $2\pi \cdot x^3 = 2\pi \cdot (t/2\pi)^3$.

When we run t from 0 to 2π in steps of 0.2 to generate $P_1(t), P_2(t)$ including the line connecting these two vertices we obtain the envelope shown by Figure 6.

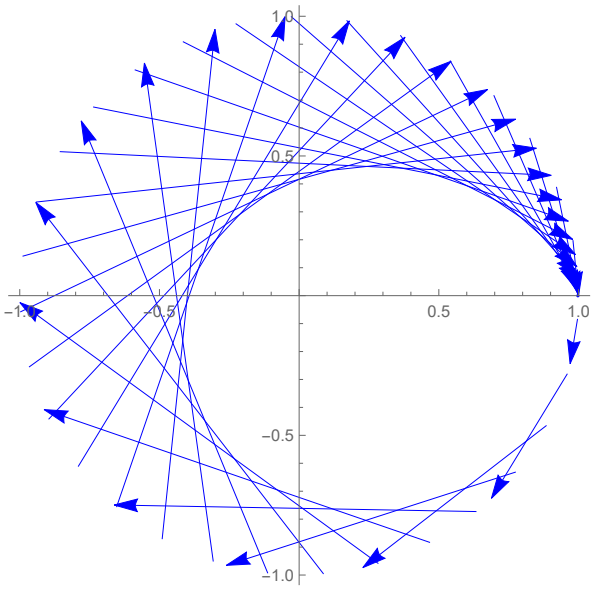


FIGURE 6. Chords from x to x^3 and their envelope

To get the equation of the envelope we insert $P_1(t)$ and $P_2(t)$ as given by equation 16 into the two-point form given by equation 2 which leads us to a parametric function $F(x,y,t)$. The envelope is given by the system of equations 1, that is $F(x,y,t) = {}^{\partial F}/_{\partial t}(x,y,t) = 0$ where:

$$F(x,y,t) = \left(\sin t - \sin \frac{t^3}{4\pi^2}\right)(x - \cos t) - \left(\cos t - \cos \frac{t^3}{4\pi^2}\right)(y - \sin t)$$

$$\frac{\partial F}{\partial t}(x,y,t) = \frac{3t^2}{4\pi^2}\left(\cos \frac{t^3}{4\pi^2}(\cos(t) - x) + \sin \frac{t^3}{4\pi^2}(\sin(t) - y)\right) - \cos\left(t - \frac{t^3}{4\pi^2}\right) + x\cos t + y \sin t$$

Solving this system for x and y gives the point of contact $P(t) = (x(t), y(t))$, in accordance with Proposition 1 for $\varphi(t) = {}^{t^3}/_{4\pi^2}$:

$$x(t) = \frac{4\pi^2 \cos\frac{t^3}{4\pi^2} + 3t^2 \cos t}{3t^2 + 4\pi^2}$$
$$y(t) = \frac{4\pi^2 \sin\frac{t^3}{4\pi^2} + 3t^2 \sin t}{3t^2 + 4\pi^2} \tag{17}$$

8. Conclusion and outlook

We have presented arithmetic envelopes as a small geometric and computational toolbox: a formula for the point of contact of a chord with its envelope, the classical epicycloids and caustics as envelopes of $x \mapsto kx \bmod 1$, a dyadic coordinate system for the odd integers in which multiplication by 3 is conjugate to the rotation by $\log_2 3$, an exact description of the region left free by its chords (Theorem 1), its heart-shaped pedal curve on the algebraic curves $x = \rho T_5(\rho)$ and $x = \rho T_7(\rho)$, and explicit implicitizations by Gröbner bases and resultants. A recurring theme is that a suitable choice of coordinates on the domain, rather than a change of the map, can turn an erratic picture into a regular one: in the dyadic labelling, the chords of $\times 3$ on every level $2^n \le x < 2^{n+1}$ belong to one and the same family of lines, and in the logarithmic labelling the convexoid becomes a disk whose radius $|\cos(\pi \log_2 3)|$ encodes the irrationality behind Furstenberg's principle. Applications of the same coordinate system to the distribution of known solutions of sums of three cubes and of cubic quadruples are discussed in [2].

Possible directions for further study include:

- the free regions of other maps and labellings, for instance of $\times p$ in base-q mantissas, where Lemma 4 applies whenever every chord has the centre on its closed major side;
- inverse problems: which families of lines, and which labellings, produce a given envelope (Appendix B)?
- other labellings, for instance by the ternary tree, by base-p mantissas or by p-adic digits, and joint pictures of $\times 2$ and $\times 3$ in the spirit of Furstenberg's principle;
- envelopes on surfaces, outer billiards, and bifurcations of families of envelopes;
- tropical and p-adic shadows of the resulting algebraic curves (Appendix A);
- juggler maps and other arithmetic maps with non-linear angle functions (Section 7.2).

Acknowledgements. The renderings in Figures 1 and 2 were produced by Max Henkel. His code for these and for the three-dimensional visualizations of [3] is available in [13].

Appendix A. Investigating caustics using tropicalization

The curves in this paper are envelopes of families of lines, and their equations quickly become unwieldy. Tropical geometry offers a coarse but robust way to look at such polynomials [23]. Tropicalization replaces a plane algebraic curve by a piecewise linear object, its tropical curve, which records which monomials of the defining polynomial dominate in which regions. Our polynomials have rational coefficients, and we compute over the field of Puiseux series with the trivial valuation on $\mathbb{Q}$. In this case the tropical curve depends only on the Newton polygon of the polynomial: it consists of rays from the origin in the directions of the outer normals of the edges of the Newton polygon, each ray weighted by the lattice length of the corresponding edge. We compute it with the library `tropical.lib` of Singular.

The red part. The Newton polygon of the red polynomial $16(x^2+y^2)^3 - 20(x^2+y^2)^2 + 5(x^2+y^2) - x$ has the vertices $(1,0)$, $(6,0)$, $(0,6)$ and $(0,2)$. Accordingly, its tropical curve (Figure 7) consists of four rays: with weight 5 downwards, with weight 4 to the left, with weight 6 in the direction $(1,1)$, and with weight 1 in the direction $(-2,-1)$.

```
1 LIB "tropical.lib";
2 ring r=(0,t),(x,y),dp;
3 poly f=16*y^6+48*x^2*y^4-20*y^4+48*x^4*y^2-40*x^2*y^2+5*y^2+16*x^6 -20*x^4+5*x^2-
      x;
4 drawTropicalCurve(f,"max");
```

Listing 1. Generate the tropical curve of the red part with Singular

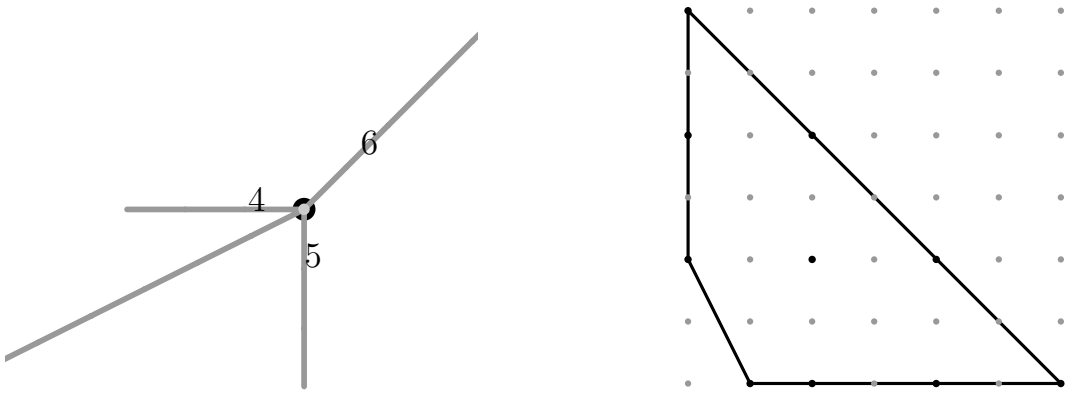


FIGURE 7. Red part tropicalized: tropical curve (left) and Newton subdivision (right)

The blue part. The Newton polygon of the blue polynomial has the vertices $(1, 0)$, $(8, 0)$, $(0, 8)$ and $(0, 2)$, so its tropical curve (Figure 8) has rays of weights 7, 6, 8 and 1 in the same four directions.

```
1 LIB "tropical.lib";
2 ring r=(0,t),(x,y),dp;
3 poly f=-64*y^8-256*x^2*y^6+112*y^6-384*x^4*y^4+336*x^2*y^4-56*y^4 -256*x^6*y
      ^2+336*x^4*y^2-112*x^2*y^2+7*y^2-64*x^8+112*x^6 -56*x^4+7*x^2+x;
4 drawTropicalCurve(f,"max");
```

LISTING 2. Generate the tropical curve of the blue part with Singular

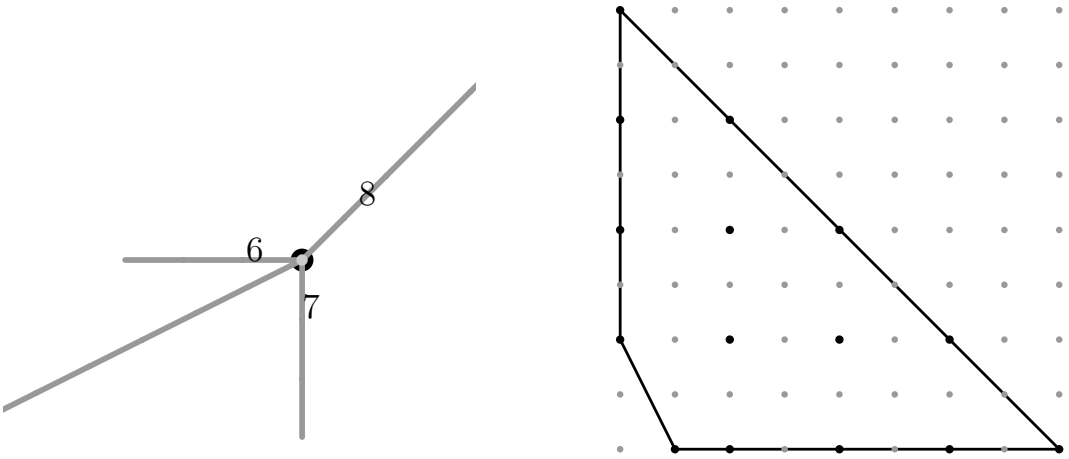


FIGURE 8. Blue part tropicalized: tropical curve (left) and Newton subdivision (right)

What the tropical picture shows. Each ray of a tropical curve corresponds to points in which the curve meets the boundary of the torus $(\mathbb{C}^*)^2$ in a suitable toric compactification, and its weight is the corresponding intersection number [23]. For the heart-shaped curve, the short edge from $(1,0)$ to $(0,2)$ records the branch $x \approx 5y^2$ resp. $x \approx -7y^2$ through the origin, i.e. the notch of the heart, while the long edges of degree 6 and 8 record the behaviour of the Chebyshev structure $x = \rho\, T_n(\rho)$ at infinity.

Outlook. Over the trivially valued field the tropical curve only sees the Newton polygon. Two variations seem worth pursuing.

- *p-adic tropicalization.* The coefficients of $\rho\, T_5(\rho)$ and $\rho\, T_7(\rho)$ have non-trivial 2-adic valuations ($16 = 2^4$, $20 = 2^2 \cdot 5$, $64 = 2^6$, $112 = 2^4 \cdot 7$, $56 = 2^3 \cdot 7$). Tropicalizing with respect to the 2-adic valuation therefore induces a non-trivial regular subdivision of the Newton polygon; for the red polynomial, for instance, the lifted vertices $(1,0)$, $(0,2)$, $(6,0)$, $(0,6)$ are not coplanar. This would connect the curves to the 2-adic nature of the dyadic labelling.
- *Families of caustics.* Families such as F_a in Section B, or caustics with a moving light source, depend on a parameter. Treating this parameter as the variable t of the Puiseux series field in the SINGULAR code above, the tropical curve records how the caustic degenerates as the parameter tends to a limit, for example when a light source moves to infinity and the cardioid turns into the nephroid.

Appendix B. A piecewise linear labelling reproducing the cardioid

The labelling of a dreamcatcher need not be linear. As an example, which is also a first step towards inverse problems for envelopes (which families of lines produce a given envelope?), we consider a piecewise linear labelling that reproduces the cardioid. We parametrize the unit circle by equation 18, so that the angle 0 is at the top and angles increase counterclockwise.

$$r(t) = (x(t), y(t)) = (-\sin t, \cos t) \tag{18}$$

Moreover we introduce two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ that are specified by equation 19.

$$f(x) = 4x \qquad g(x) = \frac{x}{2^{\lfloor \ln(x) \rfloor}} \tag{19}$$

Note that g uses the natural logarithm; we come back to the consequences of this choice below. By iterating t within an interval $]0, stop]$, we repeatedly generate a pair of lines, where the first line passes the two points P_1 and P_2 and the second line passes Q_1 and Q_2, as given by equation 20.

$$\begin{array}{ll} P_1 = (x_{P_1}, y_{P_1}) = r(2\pi \cdot g(t))) & Q_1 = (x_{Q_1}, y_{Q_1}) = r(2\pi \cdot g(t+\epsilon))) \\ P_2 = (x_{P_2}, y_{P_2}) = r(2\pi \cdot g \circ f(t)) & Q_2 = (x_{Q_2}, y_{Q_2}) = r(2\pi \cdot g \circ f(t+\epsilon)) \end{array} \tag{20}$$

Note that the operation $\circ$ is a right-to-left composition of functions, that is $g \circ f(x) = g(f(x))$. We choose a value ϵ as small as possible. As explained in Section 2, the intersection of both lines approximates a point of the envelope. Figure 9 shows such a curve – in this concrete case the variable t ran from 1 to 100 in steps of 0.2 and we chose an $\epsilon = 0.001$.

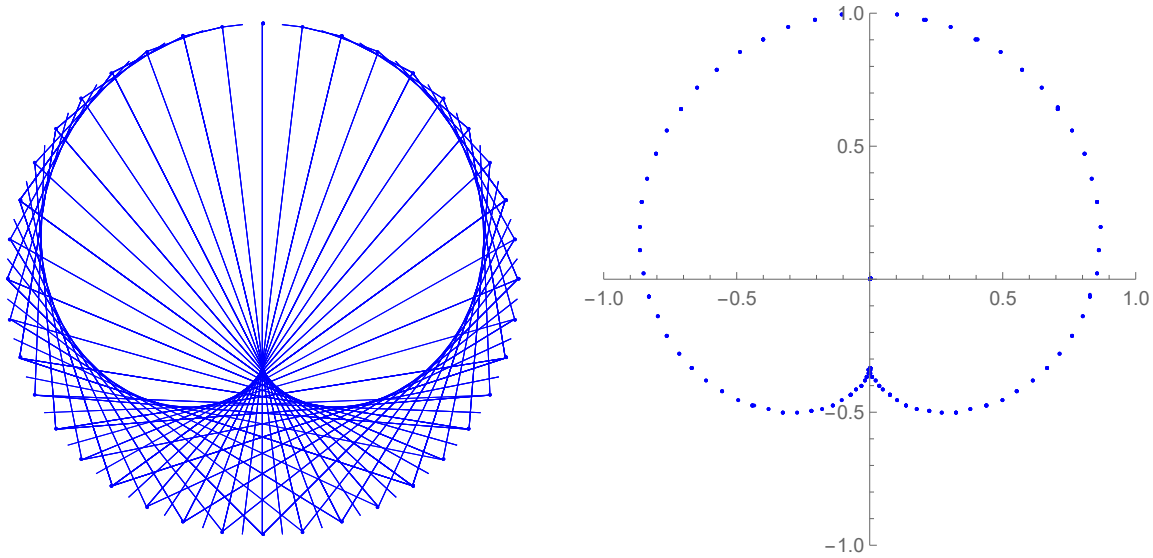


FIGURE 9. Generalized curve defined by the intersection of both lines P_1P_2 and Q_1Q_2 according to equations 20

Listing 7 in Appendix C generates this plot.
With the points of equation 20, the two-point form 2 becomes

$$F(x,y,t) = \sin\left(2\pi(g\circ f(t) - g(t))\right) - y(\sin\left(2\pi g\circ f(t)\right) - \sin\left(2\pi g(t)\right)) - x(\cos\left(2\pi g\circ f(t)\right) - \cos\left(2\pi g(t)\right)) = 0 \tag{21}$$

To obtain the equation for this curve (envelope) the system of equations 1 needs to be solved. For this we use the function 21 and determine its partial derivative with respect to t:

$$\frac{\partial F}{\partial t}(x,y,t) = 2\pi(f'(t)g'\circ f(t)\left(\cos\left(2\pi(g\circ f(t) - g(t))\right) - y\cos\left(2\pi g\circ f(t)\right) + x\sin\left(2\pi g\circ f(t)\right)\right) - g'(t)\left(\cos\left(2\pi(g\circ f(t) - g(t))\right) - y\cos\left(2\pi g(t)\right) + x\sin\left(2\pi g(t)\right)\right)) \tag{22}$$

The explicit form of both equations for f and g as in equation 19 is given in Appendix C.

Simplifying the system. We now solve the system 1 for this family. Let us generalize the function f we defined in equation 19 by substituting 4 with any $a \in \mathbb{N}$:

$$f_a(x) = ax \tag{23}$$

With regard to the function g we utilize the fact that for an integer k the truncation $\lfloor \ln(x) \rfloor$ is constant (namely, k) over successive intervals $I_k := [e^k, e^{k+1}[$. As a consequence the following equality holds for $x \in I_k$ and two functions $g_1(x) = {}^{x}/_{2^{\lfloor \ln x \rfloor}}$ and $g_2(x) = {}^{x}/_{2^k}$:

$$g_1(x) = g_2(x) \qquad g_1 \circ f_a(x) = g_2 \circ f_{a/2}(x) \tag{24}$$

Note that g_1 is our original function g as defined by equation 19. However, the second equality in 24 does not hold on all of I_k. It requires $\lfloor \ln a + \ln x \rfloor = k + 1$:

$$g_1 \circ f_a(x) = \frac{ax}{2^{\lfloor \ln a + \ln x \rfloor}} = \frac{ax}{2^{k+1}} = g_2 \circ f_{a/2}(x)$$

For $x \in I_k$ we have $k \leq \ln x < k + 1$. If $1 \leq \ln a < 2$, i.e. $e \leq a < e^2$, the equality therefore holds exactly for $\ln x < k + 2 - \ln a$, i.e. on the subinterval $[e^k, e^{k+2}/a[$ of I_k. On the remaining part $[e^{k+2}/a, e^{k+1}[$ we have $\lfloor \ln(ax) \rfloor = k + 2$ instead, and $g_1 \circ f_a(x) = ax/2^{k+2}$. For $a = 4$ this equals $g_1(x)$, so the points P_1 and P_2 coincide there and these values of t contribute no line at all. Hence, for $a = 4$, every genuine line of the family joins the angles $2\pi g_2(t)$ and $2\pi g_2 \circ f_2(t) = 2 \cdot 2\pi g_2(t)$: the family is the dreamcatcher of the doubling map, and its envelope is a cardioid, in accordance with Figure 9. Inserting into equation 21 the simpler function g_2 ($k = 1$) instead of g along with $f_{a/2}$ instead of f yields equation 25 – a greatly simplified version of equation 29. We equip the curve function F with the additional parameter a and in the given case $a = 4$:

$$F_4(x, y, t) = (y + 1)\sin(\pi t) - y\sin(2\pi t) + x\cos(\pi t) - x\cos(2\pi t) = 0 \tag{25}$$

Accordingly, the partial differential equation 30 becomes strongly simplified as well:

$$\frac{\partial F_4}{\partial t}(x, y, t) = \pi\left((y + 1)\cos(\pi t) - 2y\cos(2\pi t) - x\sin(\pi t) + 2x\sin(2\pi t)\right) = 0 \tag{26}$$

We obtain the plot given by Figure 10 by solving the system of equations made up of equations 25 and 26. Listing 8 in Appendix C generates this plot. The parameter

t ran from e to e^2. Since F_4 is 2-periodic in t, any interval of length at least 2 can be used, for example $t \in [0, 2]$.

The same curve results as a solution of the unsimplified system, equations 29 and 30 in Appendix C.

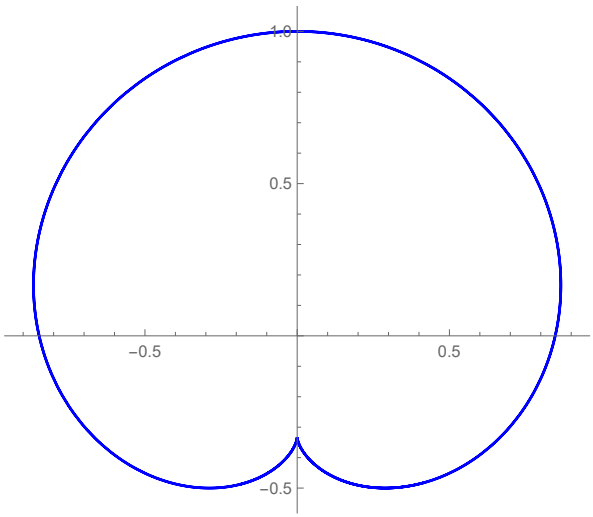


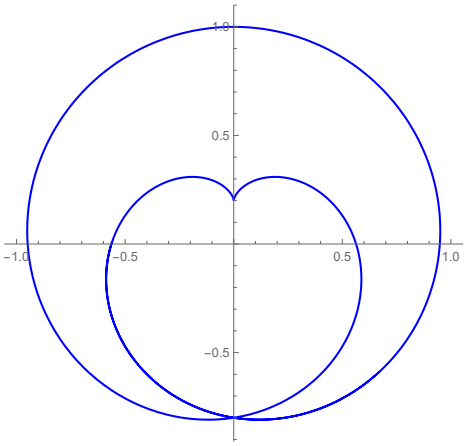
FIGURE 10. Curve defined by F_4, consistent with Figure 9, plotted here as solution of the system of equations 25 and 26

The more general function F_a involving $f_{a/2}$ instead of f is given by:

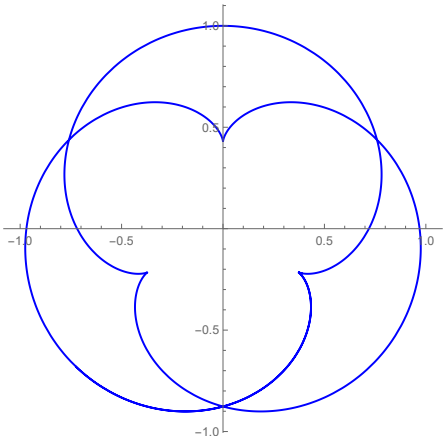
$$F_a(x, y, t) = \sin\frac{(a-2)\pi t}{2} + y\sin(\pi t) - y\sin\frac{a\pi t}{2} + x\cos(\pi t) - x\cos\frac{a\pi t}{2} = 0 \tag{27}$$

The more general partial derivative involving $f_{a/2}$ instead of f is:

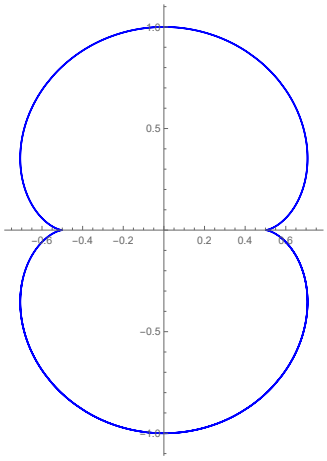
$$\frac{\partial F_a}{\partial t}(x, y, t) = \frac{\pi}{2}\left((a-2)\cos\frac{(a-2)\pi t}{2} + 2y\cos(\pi t) - ay\cos\frac{a\pi t}{2} - 2x\sin(\pi t) + ax\sin\frac{a\pi t}{2}\right) = 0 \tag{28}$$



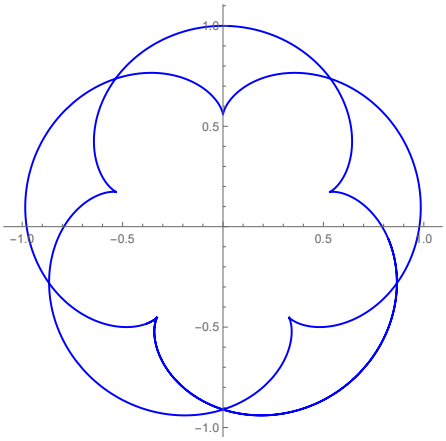
Curve F_3



Curve F_5



Curve F_6



Curve F_7

FIGURE 11. Curves F_a for $a = 3, 5, 6, 7$ as per equation 27

The curve defined by F_a is the envelope of the chords joining the angles πt and $\frac{a}{2}\pi t$, i.e. by Proposition 1 an epicycloid with ratio $\lambda = \frac{a}{2}$. Its cusps occur where the chord is a diameter, i.e. where $(\frac{a}{2} - 1)\pi t \equiv \pi \pmod{2\pi}$. For odd a the curve closes after two turns and has $a - 2$ cusps; for even a it has $\frac{a}{2} - 1$ cusps. This matches Figure 11: one cusp for F_3, three for F_5, two for F_6 (a nephroid) and five for F_7.

Appendix C. Listings and explicit formulas

Computer-assisted step in the proof of Theorem 1. With $u = e^{it/2}$ and $w = e^{it/4}$ we have $Q_r = \frac{3u^2-2u^3}{5}$ and $Q_b = \frac{3w^4-4iw^3}{7}$. On the unit circle $\overline{u} = u^{-1}$ and $\overline{w} = w^{-1}$, so an intersection point satisfies the equation $Q_r = Q_b$ and its conjugate, both polynomial after clearing denominators. The following SymPy code eliminates w by a resultant, keeps the roots on the unit circle and checks the parameter ranges $t \in [0, \frac{2\pi}{3}]$ for Q_r and $t \in [\frac{2\pi}{3}, 2\pi]$ for Q_b. Exactly one solution remains, $(t_r, t_b) \approx (1.8291729217, 2.4646905931)$. Both listings are also available as a program in [21]. The analogous computation for $Q_b(w_1) = Q_b(w_2)$, after dividing by $w_1 - w_2$, has no solution with $w_1 \neq w_2$ in the parameter range of Q_b.

```
import sympy as sp, numpy as np
u, w = sp.symbols('u w'); I = sp.I
P1 = sp.expand(7*(3*u**2 - 2*u**3) - 5*(3*w**4 - 4*I*w**3))  # Q_r = Q_b
P2 = sp.expand(7*(3*u - 2)*w**4 - 5*(3 + 4*I*w)*u**3)        # conjugate
sols = []
for f, _ in sp.factor_list(sp.resultant(P1, P2, w), extension=I)[1]:
    if sp.Poly(f, u).degree() == 0: continue
    for r in sp.Poly(f, u).nroots(n=40, maxsteps=500):
        r = complex(r)
        if abs(abs(r) - 1) > 1e-12: continue
        for s in sp.Poly(P1.subs(u, r), w).nroots(n=40, maxsteps=500):
            s = complex(s)
            if abs(abs(s) - 1) < 1e-10 and abs(complex(P2.subs({u: r, w: s}))) <
    1e-9:
                tr, tb = 2*(np.angle(r) % (2*np.pi)), 4*(np.angle(s) % (2*np.pi))
                if tr <= 2*np.pi/3 and tb >= 2*np.pi/3: sols.append((tr, tb))
print(sols)   # [(1.8291729217..., 2.4646905931...)]
```

Listing 3. Intersections of the arcs of Theorem 1

```
w1, w2 = sp.symbols('w1 w2')
f = sp.cancel((3*(w1**4 - w2**4) - 4*I*(w1**3 - w2**3)) / (w1 - w2))
g = sp.cancel(sp.expand((3*(w1**-4 - w2**-4) + 4*I*(w1**-3 - w2**-3))
                        * w1**4 * w2**4) / (w2 - w1))
inr = lambda x: 2*np.pi/3 - 1e-9 <= x <= 2*np.pi + 1e-9
bad = []
for q, _ in sp.factor_list(sp.resultant(f, g, w2), extension=I)[1]:
    if sp.Poly(q, w1).degree() == 0: continue
    for r in sp.Poly(q, w1).nroots(n=40, maxsteps=500):
        r = complex(r)
```

```
        if abs(abs(r) - 1) > 1e-10: continue
        for s in sp.Poly(f.subs(w1, r), w2).nroots(n=40, maxsteps=500):
            s = complex(s)
            if abs(abs(s) - 1) < 1e-10 and abs(complex(g.subs({w1: r, w2: s}))) <
    1e-8:
                a, b = 4*(np.angle(r) % (2*np.pi)), 4*(np.angle(s) % (2*np.pi))
                if inr(a) and inr(b) and abs(a - b) > 1e-7: bad.append((a, b))
print(bad)    # []
```

LISTING 4. The arc Q_b has no self-intersection

```
F[x_, y_, z_] := -(a + 1) (x + I*y) + a*z + z^a;
G[x_, y_, z_] := -(a + 1) (x - I*y) z^a + a*z^(a - 1) + 1;
det = Resultant[F[x, y, z], G[x, y, z], z]/(a + 1)^(a - 1)/(-4)^(Mod[a, 2]);
cl = CoefficientList[det, {x, y}];
Poly[x_, y_] := Internal`FromCoefficientList[cl, {x, y}];
```

LISTING 5. Listing for implicitizing the curve given by parameterized equations 14

The piecewise linear labelling of Section B. Inserting f and g from equation 19 into equation 21 gives

$$F(x,y,t)=\sin\left(\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}-\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}\right)-y\left(\sin\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}-\sin\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}\right)-x\left(\cos\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}-\cos\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}\right)=0 \tag{29}$$

and inserting them into equation 22 gives

$$\frac{\partial F}{\partial t}(x,y,t)=2\pi\left(\frac{4}{2^{\lfloor \ln 4t\rfloor}}\left(\cos\left(\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}-\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}\right)-y\cos\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}+x\sin\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}\right)-\frac{1}{2^{\lfloor \ln t\rfloor}}\left(\cos\left(\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}-\frac{2\pi 4t}{2^{\lfloor \ln 4t\rfloor}}\right)-y\cos\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}+x\sin\frac{2\pi t}{2^{\lfloor \ln t\rfloor}}\right)\right) \tag{30}$$

The parametric vector equation [24, p. 218] for this line passing through P_1 and P_2 has the form:

$$\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x_{P_1}\\y_{P_1}\end{pmatrix}+u\begin{pmatrix}x_{P_2}-x_{P_1}\\y_{P_2}-y_{P_1}\end{pmatrix} \tag{31}$$

This vector equation of the line through P_1 and P_2 of Section B can be used directly to generate a parametric plot, see Listing 6 and Figure 12.

```
r[t_] := (x = -Sin[t]; y = Cos[t]; {x, y});
f[x_] := x*4;
g[x_] := x/2^Floor[Log[x]];
parametricFunc[t_, u_] := (P1 = r[2 Pi*g[t]];
   P2 = r[2 Pi*g[f[t]]]; {P1[[1]] + u*(P2[[1]] - P1[[1]]),
    P1[[2]] + u*(P2[[2]] - P1[[2]])});
ParametricPlot[parametricFunc[t, u], {t, 0.01, 1}, {u, 0, 1},
 PlotStyle -> Blue]
```

LISTING 6. Listing for the parametric plot shown by Figure 12

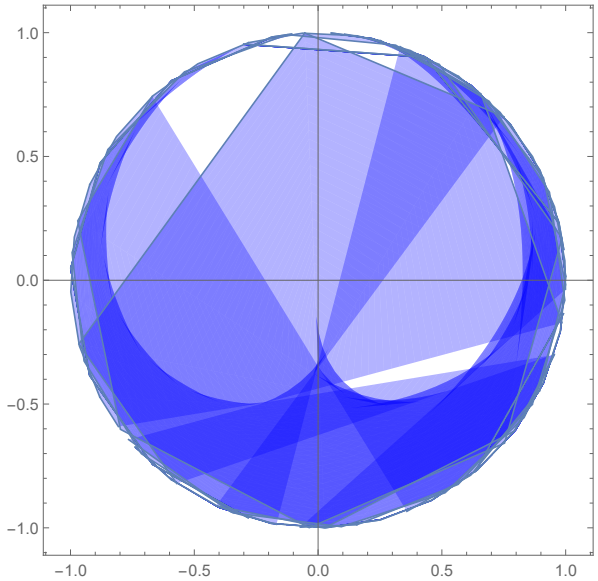


FIGURE 12. Parametric plot of the generalized curve defined by the parametric vector equation 31

```
r[t_] := (x = -Sin[t]; y = Cos[t]; {x, y});

intersectLines[line1_, line2_] := (
   a11 = line1[[1, 2]] - line1[[2, 2]];
   a12 = line1[[2, 1]] - line1[[1, 1]]; det1 = Det[line1];
   a21 = line2[[1, 2]] - line2[[2, 2]];
   a22 = line2[[2, 1]] - line2[[1, 1]]; det2 = Det[line2];
```

```
   LinearSolve[{{a11, a12}, {a21, a22}}, {-det1, -det2}]);

generatePlot[f_, g_, step_] := (
   start = 1;
   stop = 100;
   e = 0.001;
   lines1 =
    Table[{r[2 Pi*g[t]], r[2 Pi*g[f[t]]]}, {t, start, stop,
      step}];
   lines2 =
    Table[{r[2 Pi*g[t + e]], r[2 Pi*g[f[t + e]]]}, {t,
      start, stop, step}];
   plot1 =
    Graphics[{Blue, Thickness[0.0001], Map[Line, lines1],
      PlotRange -> {{-1, 1}, {-1, 1}}, AspectRatio -> 1}];
   plot2 =
    ListPlot[MapThread[intersectLines, {lines1, lines2}],
     PlotRange -> {{-1, 1}, {-1, 1}}, AspectRatio -> 1,
     PlotStyle -> Blue];
   GraphicsGrid[{{plot1, plot2}}]
   );

f[x_] := x*4;
g[x_] := x/2^Floor[Log[x]];
generatePlot[f, g, 0.2]
```

LISTING 7. Listing for the plot shown by Figure 9

```
F[x_, y_, t_] := (y + 1) Sin[Pi*t] - y*Sin[2 Pi*t]
                     + x*Cos[Pi*t] - x*Cos[2 Pi*t];
Fd[x_, y_, t_] := (y + 1) Cos[Pi*t] - 2 y*Cos[2 Pi*t]
                      - x*Sin[Pi*t] + 2 x*Sin[2 Pi*t];
parametricFunc[t_] := {x, y} /. (Solve[F[x, y, t] == 0
                                    && Fd[x, y, t] == 0, {x, y}]);
ParametricPlot[parametricFunc[t], {t, E, E^2}, PlotStyle -> Blue]
```

LISTING 8. Listing for the plot shown by Figure 10

Appendix D. Algebraic background

For the convenience of the reader we recall the notions behind the Prüfer group and p-adic numbers used in Section 4.

cosets	Let G be a group and $H \subset G$ a subgroup. A left coset of H in G is a subset of G of the form $aH := \{ah \mid h \in H\}$ where $a \in G$; the set of left cosets is denoted by G/H. Analogously, a right coset is a set $Ha := \{ha \mid h \in H\}$; the set of right cosets is denoted by $H\backslash G$, where $a \in G$ [25, p. 33], [26, p. 20-21]. In the case of additively written groups, cosets are also written as $a+H$ and $H+a$ [25, p. 33].
normal subgroup	A subgroup $H \subset G$ is called a "normal subgroup" if $aH = Ha$ holds for all $a \in G$ [25, p. 38], [26, p. 21], [27, p. 188]. The coset aH or Ha belonging to a is referred to as the residue class of a modulo H [26, p. 21].
quot. group	Let G be a group and $N \lhd G$ a normal subgroup. The set of the (left or right, it doesn't matter) cosets of N in G together with the multiplication $gN \cdot hN := ghN$ forms the quotient group (factor group) of N in G, written G/N and read "G modulo N" [27, p. 200], [25, p. 40].
quot. group $\mathbb{Q}/\mathbb{Z}$	The group of rationals modulo one $\mathbb{Q}/\mathbb{Z} = (\mathbb{Q}, +)/\langle 1\rangle = \{q + \mathbb{Z} \mid q \in \mathbb{Q}\}$ contains the cosets of $\mathbb{Z}$. $\mathbb{Z}$ is the identity of $\mathbb{Q}/\mathbb{Z}$ and it is a normal subgroup (generated by the element 1) of $\mathbb{Q}$. The cosets are residue classes created by the congruence relation $a \sim b \Leftrightarrow a-b \in \mathbb{Z}$, see [28, p. 27].
group $\mathbb{Z}[1/p]$	Let p be a prime. The group $\mathbb{Z}[1/p]$ denotes the additive (sub)group of rational numbers whose denominator is a power of p.
Prüfer group $\mathbb{Z}(p^\infty)$	The Prüfer p-group is the group of rationals whose denominator is a power of p, modulo one: $\mathbb{Z}(p^\infty) = \mathbb{Z}[1/p]/\mathbb{Z}$, see [29, p. 16], [30, p. 8]. We can also write $\mathbb{Z}(p^\infty) = \mathbb{Q}_p/\mathbb{Z}_p$, where $\mathbb{Q}_p$ denotes the additive group of (the field of) p-adic numbers and $\mathbb{Z}_p$ the additive group of (the ring of) p-adic integers.

Cauchy seq. in $\mathbb{Q}$, $\mathbb{R}$	A sequence $(x_i)_{i\in\mathbb{N}}$ in $\mathbb{Q}$ or $\mathbb{R}$ is a Cauchy sequence if for any $\epsilon > 0$ there exists a positive integer N such that $\lvert x_n - x_m\rvert < \epsilon$ for all $n, m \geq N$, see [31, p. 153], [32, p. 24], [33, p. 10].
p-adic val. for $\mathbb{Z}$	Fix a prime number p in $\mathbb{Z}$. The p-adic valuation of a nonzero integer $n = r \cdot p^{v_p(n)}$ is the highest exponent $v_p(n)$ such that $p^{v_p(n)}$ divides n (we say $p^{v_p(n)}$ divides n "exactly"). Hence p and r are coprime. If n, p are coprime then $v_p(n) = 0$, and by convention $v_p(0) = \infty$, see [34].
p-adic val. for $\mathbb{Q}$	The p-adic valuation can be extended to the field of rational numbers. Let $x = n \cdot s^{-1}$ be a rational number, then $v_p(x) = v_p(n) - v_p(s)$. Any nonzero rational number x can be uniquely represented as $x = rp^{v_p(x)}s^{-1}$, where $r, s \in \mathbb{Z}$, $s > 0$, and $\gcd(r, s) = \gcd(r, p) = \gcd(s, p) = 1$, see [31, p. 154], [35].
p-adic norm	Let x be any number in $\mathbb{Q}$, for which we already know that it can be written as $x = rp^{v_p(x)}s^{-1}$, where p is a prime number, $s > 0$ and r are integers not divisible by p. The p-adic norm of x is defined by $\lvert x\rvert_p = p^{-v_p(x)}$ for $x \neq 0$, and $\lvert 0\rvert_p = 0$, see [34], [31, p. 154], [36].
p-adic dist.	Let $x, y \in \mathbb{Q}$. The p-adic distance between x and y is defined by $d_p(x, y) = \lvert x - y\rvert_p$, see [31, p. 155].
$\mathbb{Q}_p$	The field $\mathbb{Q}_p$ of p-adic numbers is the set of equivalence classes of Cauchy sequences [33, p. 10]. The elements of $\mathbb{Q}_p$, the so-called p-adic numbers, are equivalence classes of Cauchy sequences $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ with respect to the equivalence relation $(a_n) \sim (b_n)$ if $(a_n - b_n)$ is a p-adic zero sequence, see [31, p. 159]. Furthermore $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ with respect to the p-adic distance d_p [31, p. 159].
$\mathbb{Z}_p$	The ring $\mathbb{Z}_p$ of p-adic integers is the completion of $\mathbb{Z}$ with respect to the p-adic norm. That is, $\mathbb{Z}_p$ is the set of all equivalence classes of Cauchy sequences (a_n) where (a_n) and (b_n) are equivalent if $\lim_{n\to\infty}\lvert a_n - b_n\rvert_p = 0$, see [37]. $\mathbb{Z}_p$ is a local ring whose maximal ideal is the principal ideal $p\mathbb{Z}_p = \{x \in \mathbb{Q}_p : \lvert x\rvert_p < 1\}$, see [38, p. 74]. The elements of $\mathbb{Z}_p$ correspond to the infinite series $\sum_{v=0}^{\infty} r_v p^v$ with integer coefficients $0 \leq r_v \leq p - 1$ [26, p. 203, 205].

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